

Introduction to Machine Learning

Topics:

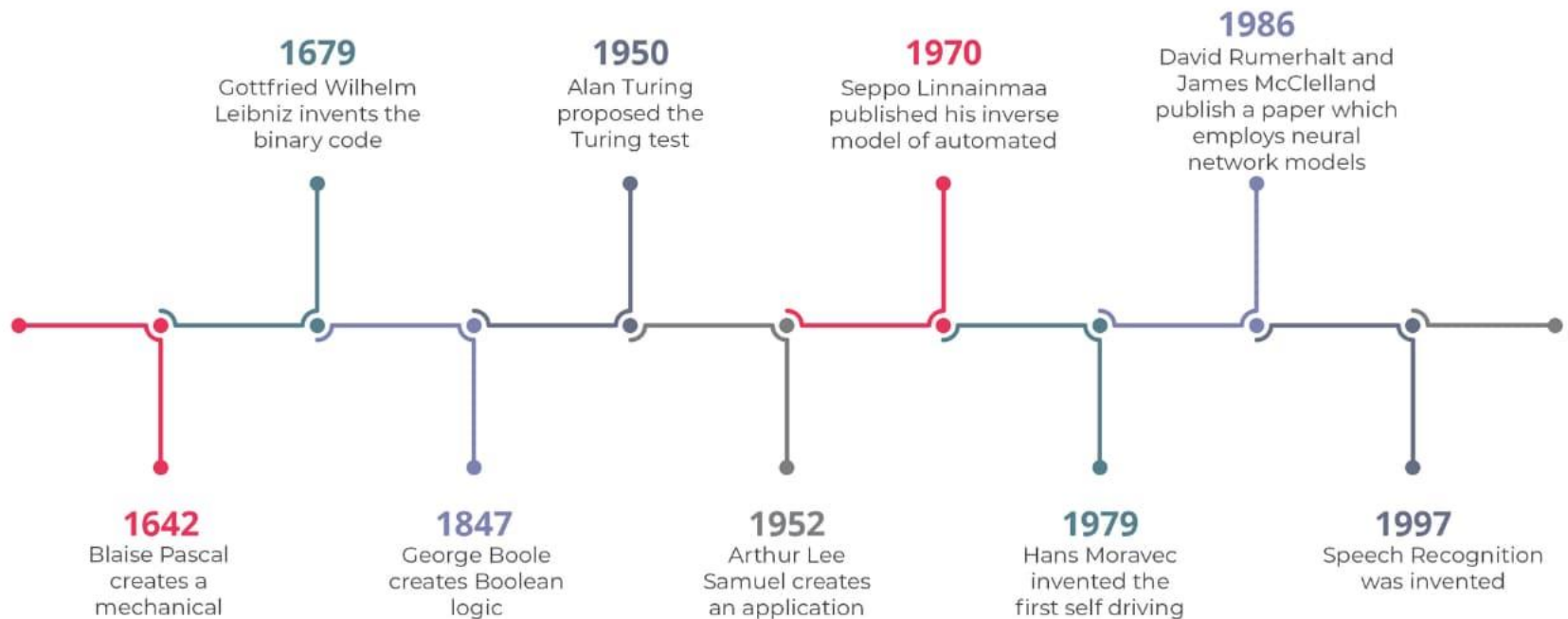
- Evolution of Machine Learning
- Paradigms for Machine Learning
- Learning by Rote
- Learning by Induction

What is Machine Learning?

- Machine Learning (ML) is a field of AI
- Systems learn patterns from data
- Improves performance without explicit programming
- **Simple idea:**
- Data → Learning Algorithm → Model → Predictions

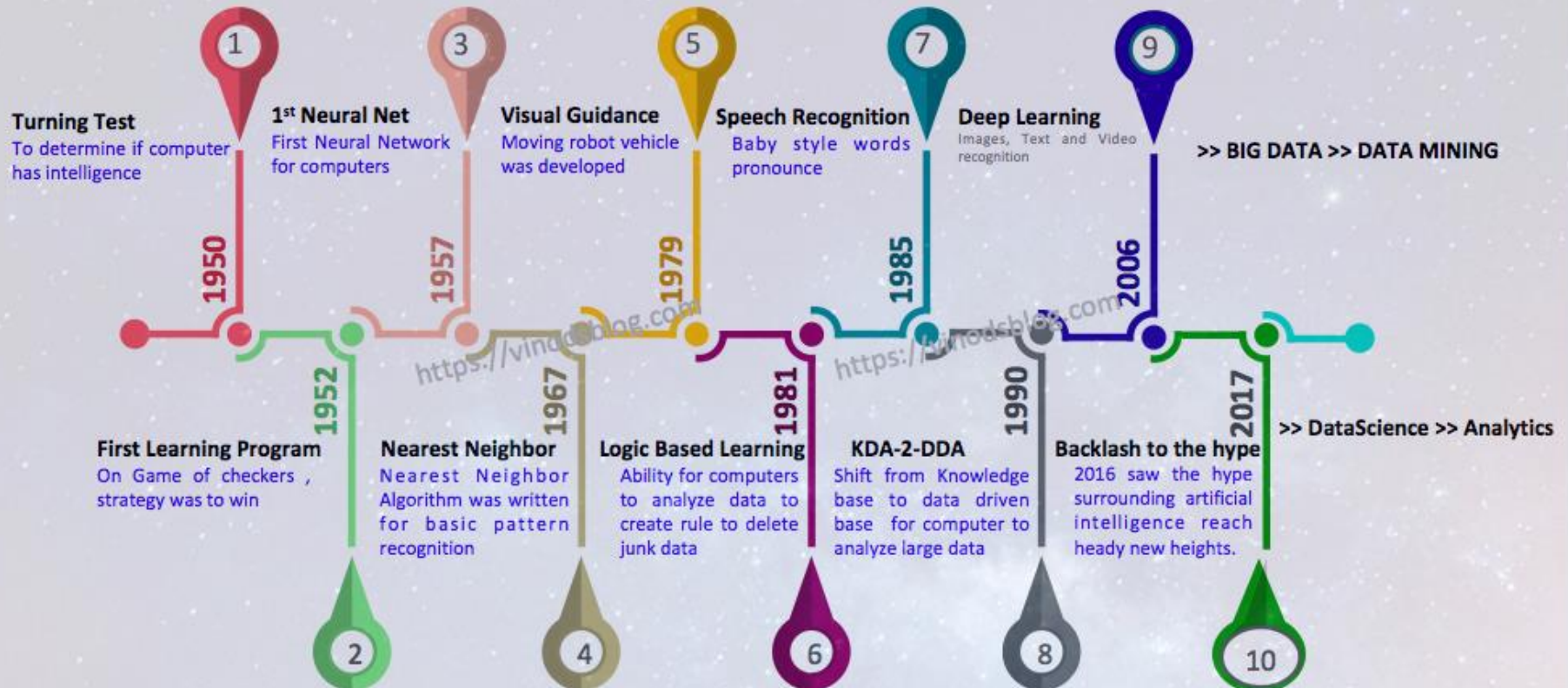
Evolution of Machine Learning

DETAILED HISTORY OF MACHINE LEARNING



Evolution of Machine Learning

Machine Learning Evolution Over The Years



Data Source - Open Internet various sources

Image Source - <https://vinodsblog.com>

via @vinod1975

Evolution of Machine Learning

A.I. TIMELINE

SYZYG

1950

TURING TEST

Computer scientist Alan Turing proposes a test for machine intelligence. If a machine can trick humans into thinking it is human, then it has intelligence

1955

A.I. BORN

Term 'artificial intelligence' is coined by computer scientist, John McCarthy to describe "the science and engineering of making intelligent machines"

1961

UNIMATE

First industrial robot, Unimate, goes to work at GM replacing humans on the assembly line

1964

ELIZA

Pioneering chatbot developed by Joseph Weizenbaum at MIT holds conversations with humans

1966

SHAKY

The 'first electronic person' from Stanford, Shakey is a general-purpose mobile robot that reasons about its own actions

A.I.
WINTER

Many false starts and dead-ends leave A.I. out in the cold

1997

DEEP BLUE

Deep Blue, a chess-playing computer from IBM defeats world chess champion Garry Kasparov

1998

KISMET

Cynthia Breazeal at MIT introduces Kismet, an emotionally intelligent robot insofar as it detects and responds to people's feelings



1999

AIBO

Sony launches first consumer robot pet dog AiBO (AI robot) with skills and personality that develop over time



2002

ROOMBA

First mass produced autonomous robotic vacuum cleaner from iRobot learns to navigate and clean homes



2011

SIRI

Apple integrates Siri, an intelligent virtual assistant with a voice interface, into the iPhone 4S



2011

WATSON

IBM's question answering computer Watson wins first place on popular \$1M prize television quiz show Jeopardy



2014

EUGENE

Eugene Goostman, a chatbot passes the Turing Test with a third of judges believing Eugene is human



2014

ALEXA

Amazon launches Alexa, an intelligent virtual assistant with a voice interface that completes shopping tasks



2016

TAY

Microsoft's chatbot Tay goes rogue on social media making inflammatory and offensive racist comments



2017

ALPHAGO

Google's A.I. AlphaGo beats world champion Ke Jie in the complex board game of Go, notable for its vast number (2^{170}) of possible positions

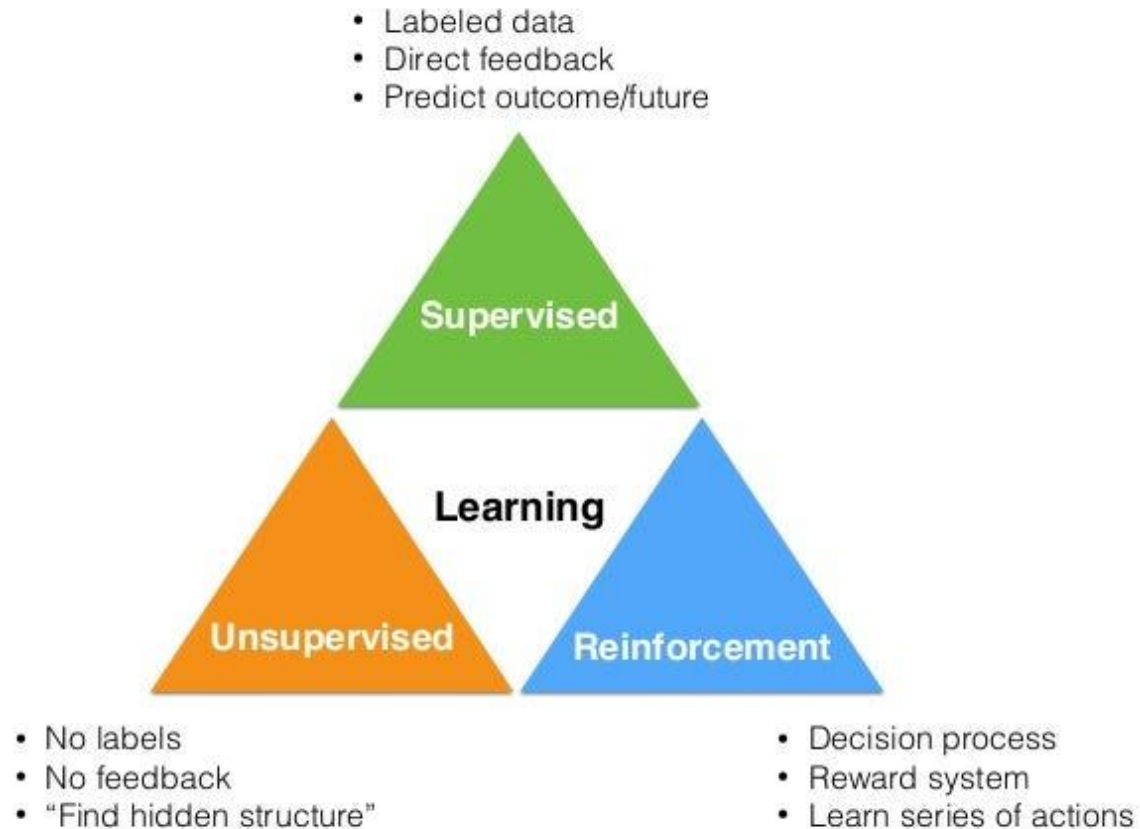
Evolution of Machine Learning

- **Key Stages:**
- **1950s–60s:** Rule-based AI, Perceptrons
- **1970s–80s:** Expert systems
- **1990s:** Statistical learning, decision trees
- **2000s:** Big data & ensemble methods
- **2010s–Present:** Deep learning & neural networks

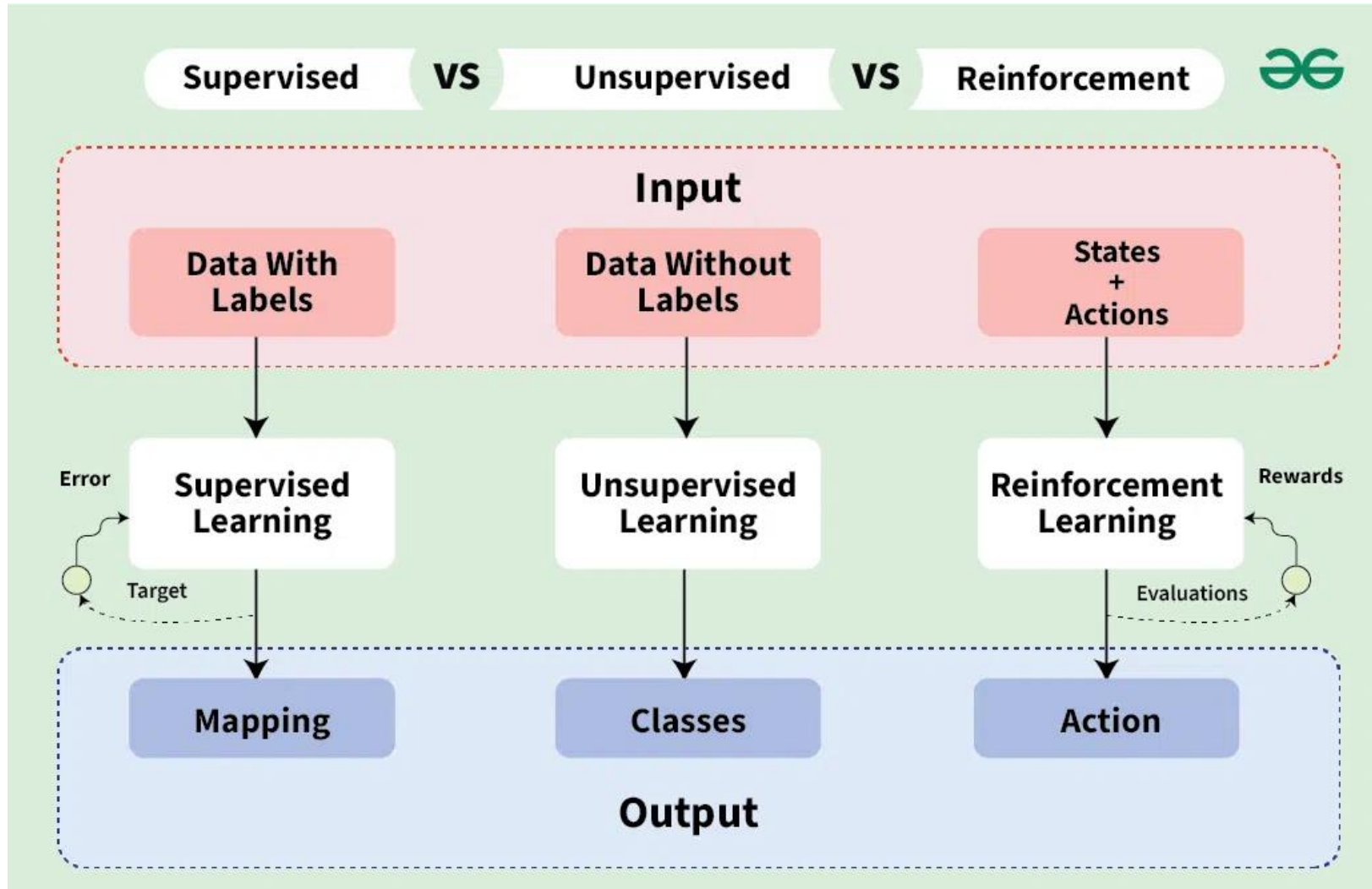
Evolution Timeline Diagram

- Rule-Based AI
- ↓
- Statistical Learning
- ↓
- Machine Learning
- ↓
- Deep Learning
- Trend:
- More data + more computation → better learning capability

Paradigms for Machine Learning



Paradigms for Machine Learning



Paradigms for Machine Learning

- **Main Paradigms:**
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Semi-supervised Learning

Paradigms Explained (with Table)

Paradigm	Data Type	Example
Supervised	Labeled	Spam detection
Unsupervised	Unlabeled	Customer clustering
Reinforcement	Reward-based	Game playing AI
Semi-supervised	Partial labels	Image classification

Learning by Rote

Definition:

- Learning by memorization
- No generalization
- Stores exact examples
- **Example:**
- Input: $(2 + 3)$ Stored Answer: 5

If input changes $\rightarrow$ system may fail

Learning by Rote

Experience



Memory Storage



Exact Recall

Learning by Rote

Characteristics:

- Very fast
- No reasoning
- Limited usefulness

Learning by Induction

- **Definition:**
- Learns general rules from examples
- Can handle unseen data
- Most common ML approach

Learning by Induction

Example:

$$2+3=5$$

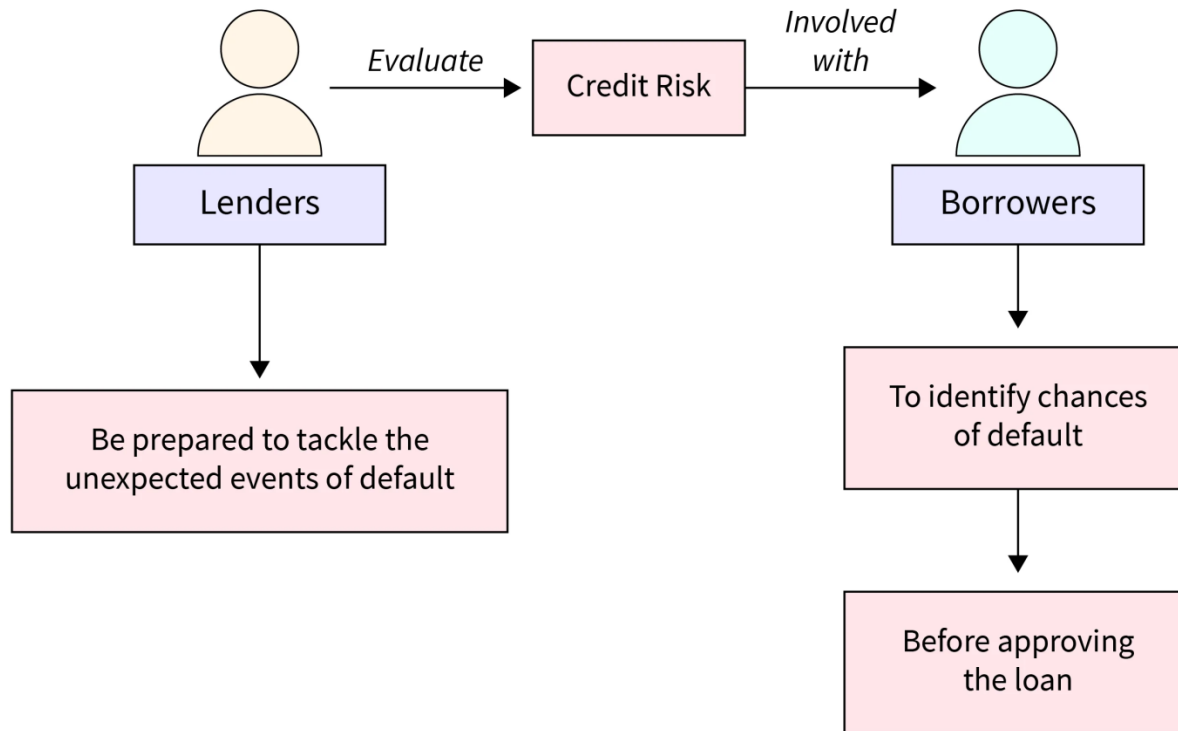
$$4+3=7$$

$$6+3=9$$

Induced Rule:

$$x + 3$$

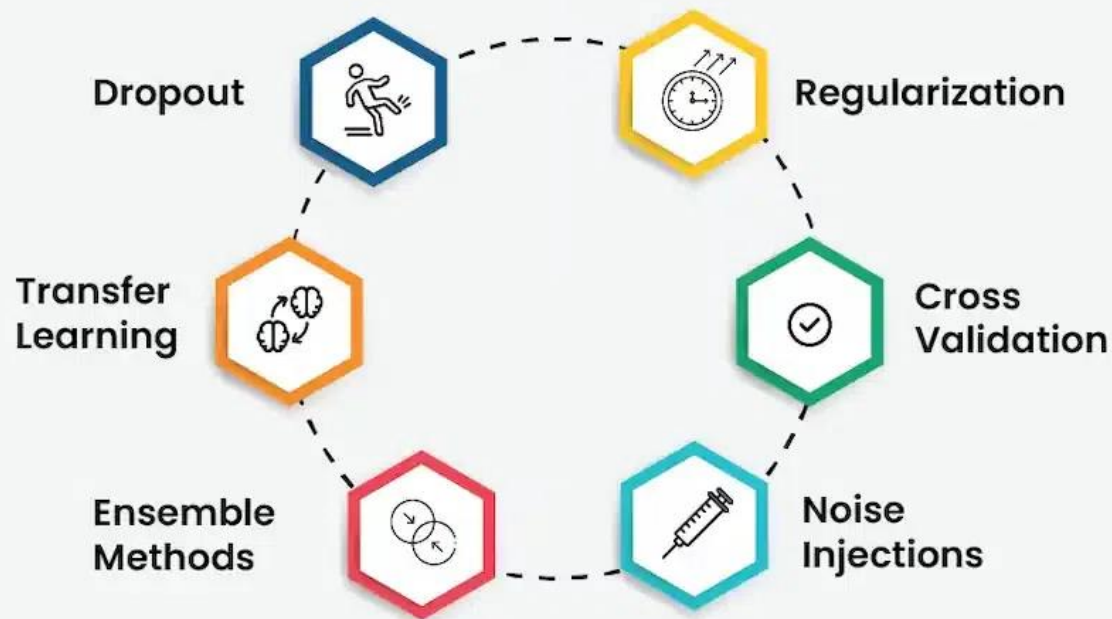
Learning by Induction



Learning by Induction



Generalization Rules & Techniques in AI



Learning by Induction

Examples



Pattern Detection



General Rule

Rote vs Induction

Feature	Rote Learning	Inductive Learning
Memory	High	Moderate
Generalization	 No	 Yes
Flexibility	Low	High
ML Usage	Rare	Very common

Reinforcement Learning (RL)

- A Machine Learning Paradigm
- Learning through interaction and rewards

What is Reinforcement Learning?

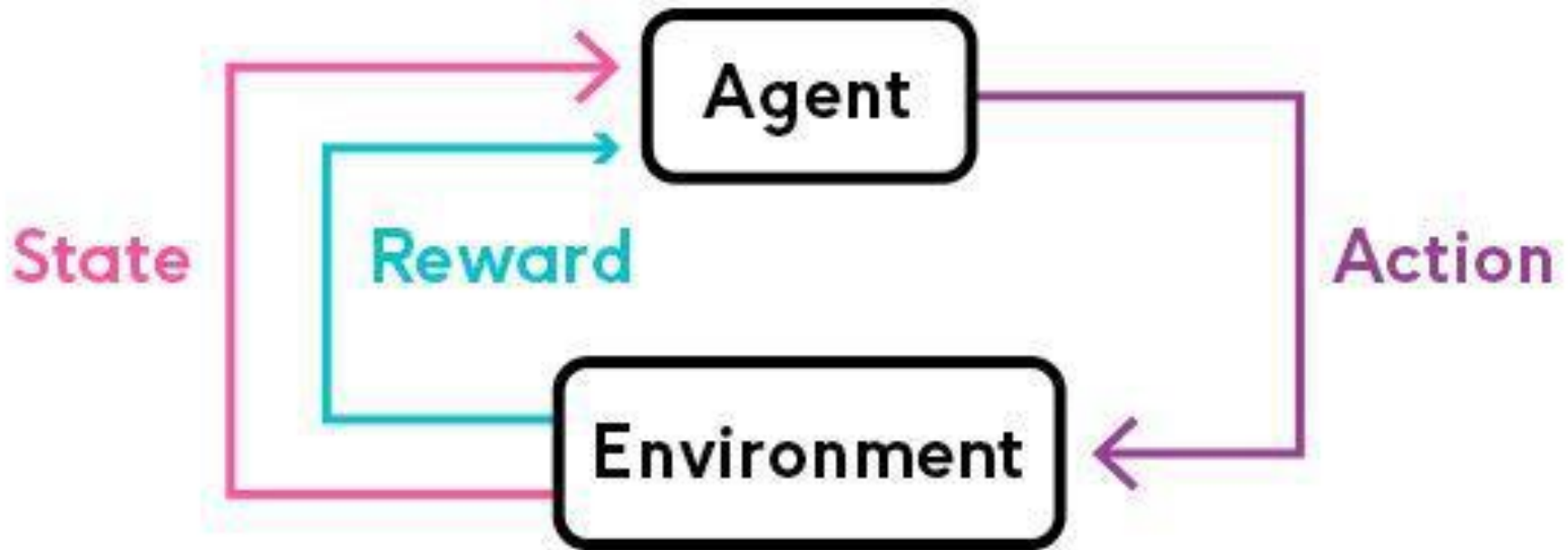
Definition:

Reinforcement Learning is a type of Machine Learning where an **agent learns by interacting with an environment and receiving rewards or penalties.**

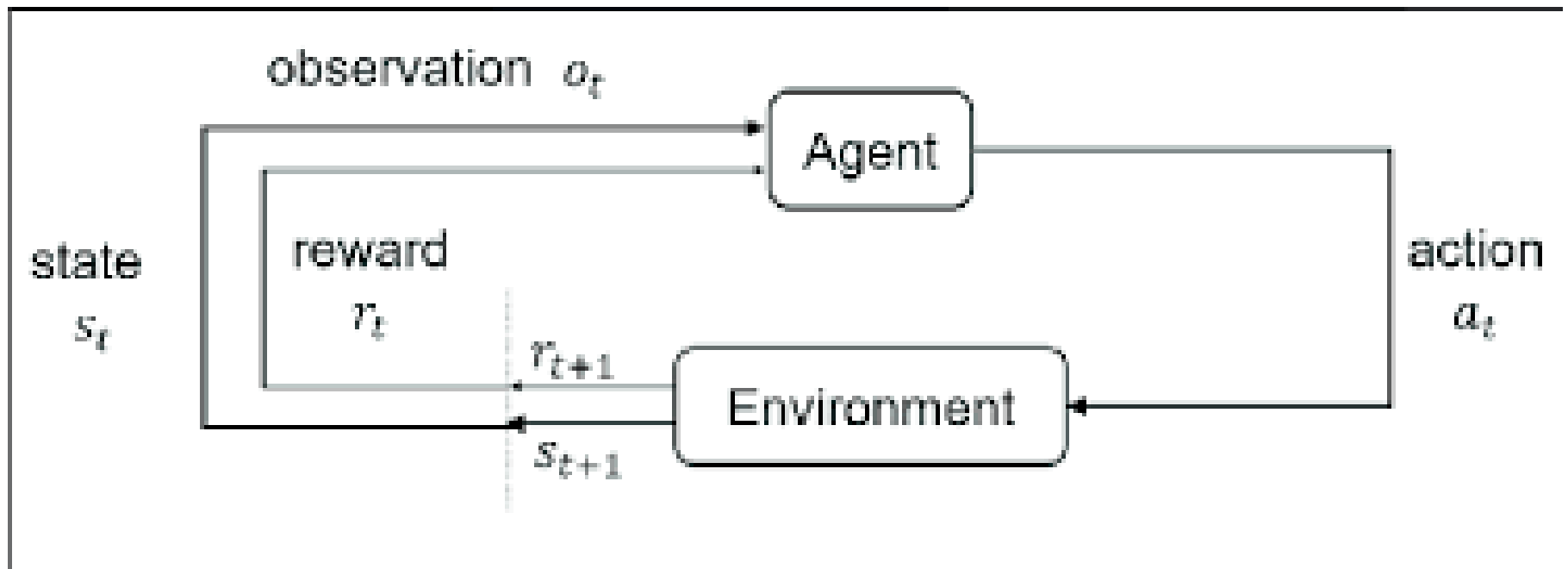
Goal:

 Maximize cumulative reward over time

Key Components of Reinforcement Learning



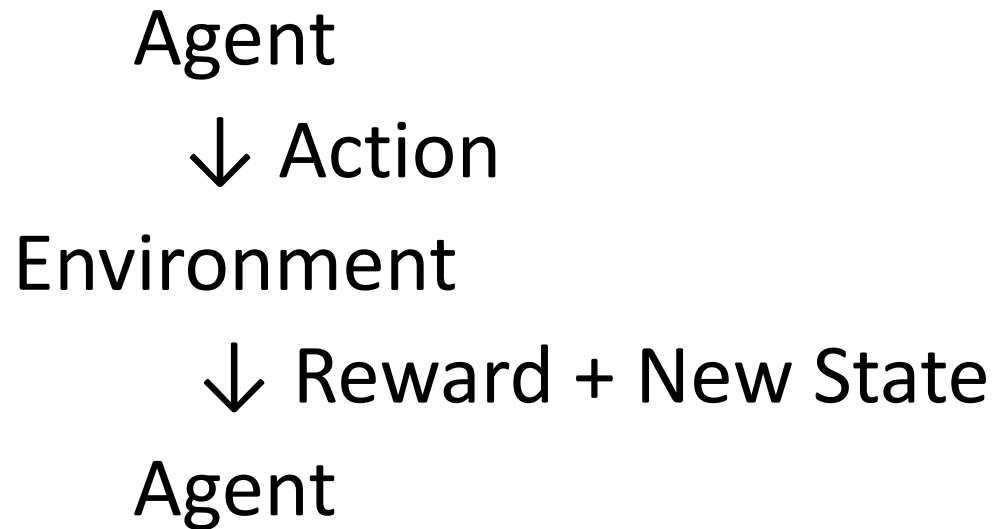
Key Components of Reinforcement Learning



Key Components of Reinforcement Learning

- **Agent** – Learner or decision maker
- **Environment** – Everything the agent interacts with
- **Action (A)** – What the agent does
- **State (S)** – Current situation
- **Reward (R)** – Feedback from environment

Reinforcement Learning Cycle



- Learning happens through continuous interaction

How Reinforcement Learning Works

1. Agent observes the current state
2. Agent selects an action
3. Environment responds with:
 - a) New state
 - b) Reward
4. Agent updates its strategy
5. Process repeats

Example of Reinforcement Learning

			end +1
			end -1
start			

Example of Reinforcement Learning

Robot in a Maze

- Goal: Reach destination
- Reward: +10 for success
- Penalty: -1 for hitting walls

Example of Reinforcement Learning

0 START	6	12	18	24	30	36	42
1	7	13	19	25	31	37	43
2	8	14	20	26	32	38	44
3	9	15	21	27	33	39	45
4	10	16	22	28	34	40	46
5	11	17	23	29	35	41	47 END

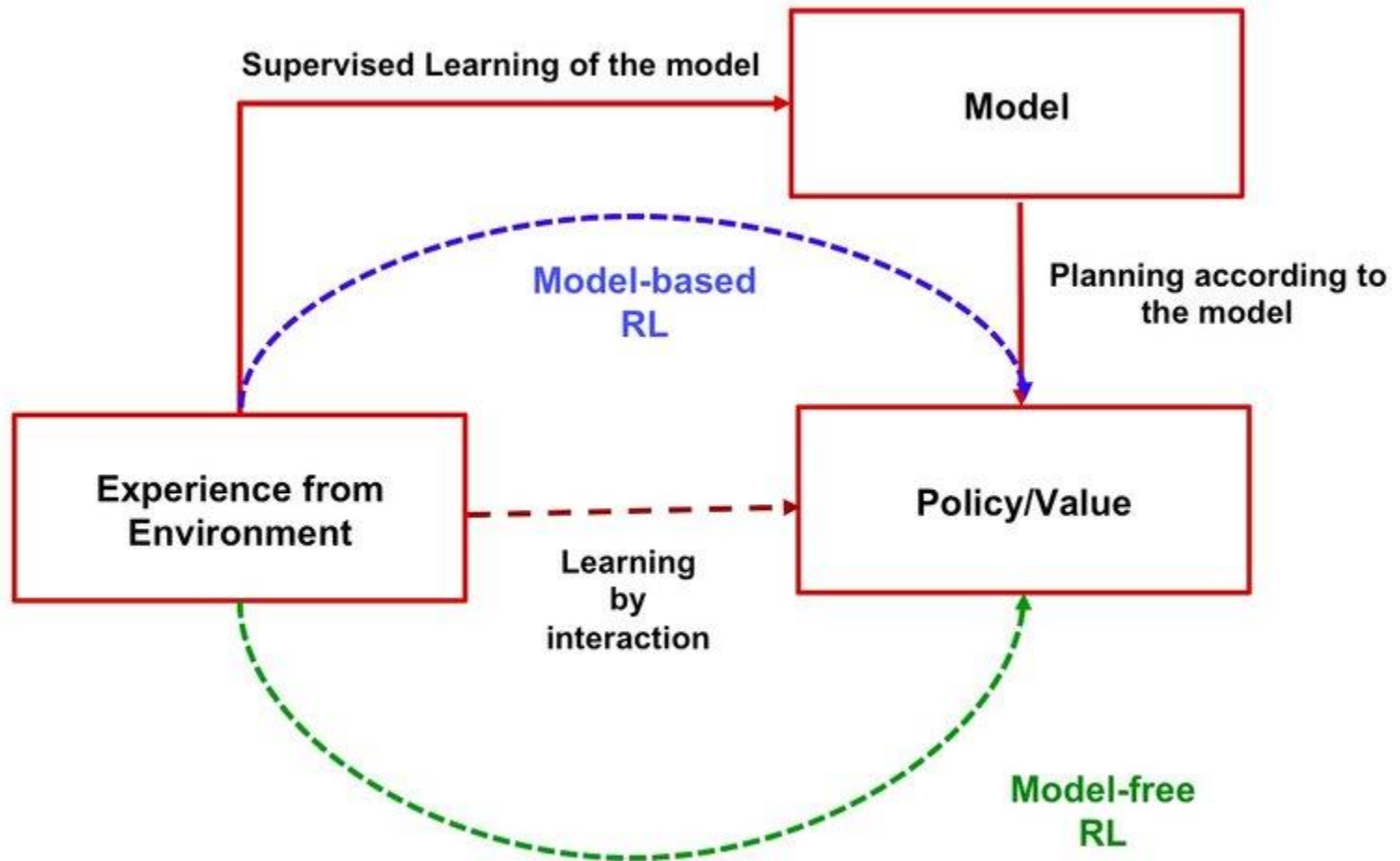
Policy in Reinforcement Learning

- **Policy (π):**
- A strategy used by the agent
- Maps states $\rightarrow$ actions
- **Example:**
- If obstacle ahead $\rightarrow$ turn left Else $\rightarrow$ move forward

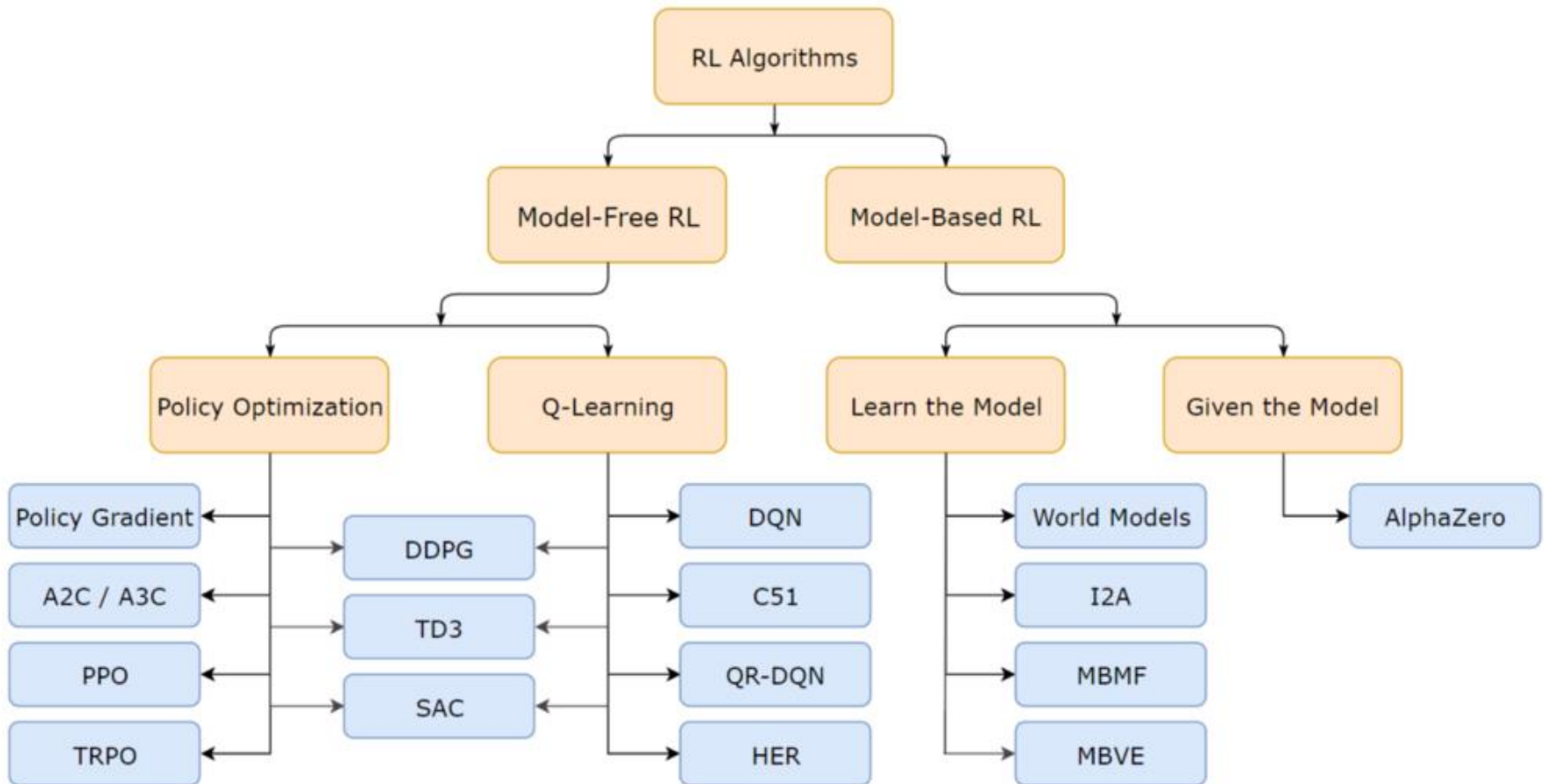
Reward Function

- Defines what is **good or bad behavior**
- Guides the learning process
- **Example Rewards:**
- Win a game $\rightarrow +100$
- Lose a game $\rightarrow -100$
- Each move $\rightarrow -1$

Types of Reinforcement Learning



Types of Reinforcement Learning



Types of Reinforcement Learning

- **Model-Based RL**
- Environment is known
- **Model-Free RL**
- Environment is unknown (most common)

Exploration vs Exploitation

Exploration:

- Trying new actions
- Discover better rewards

Exploitation:

- Using known actions
- Maximizing reward
- **Balance is important**

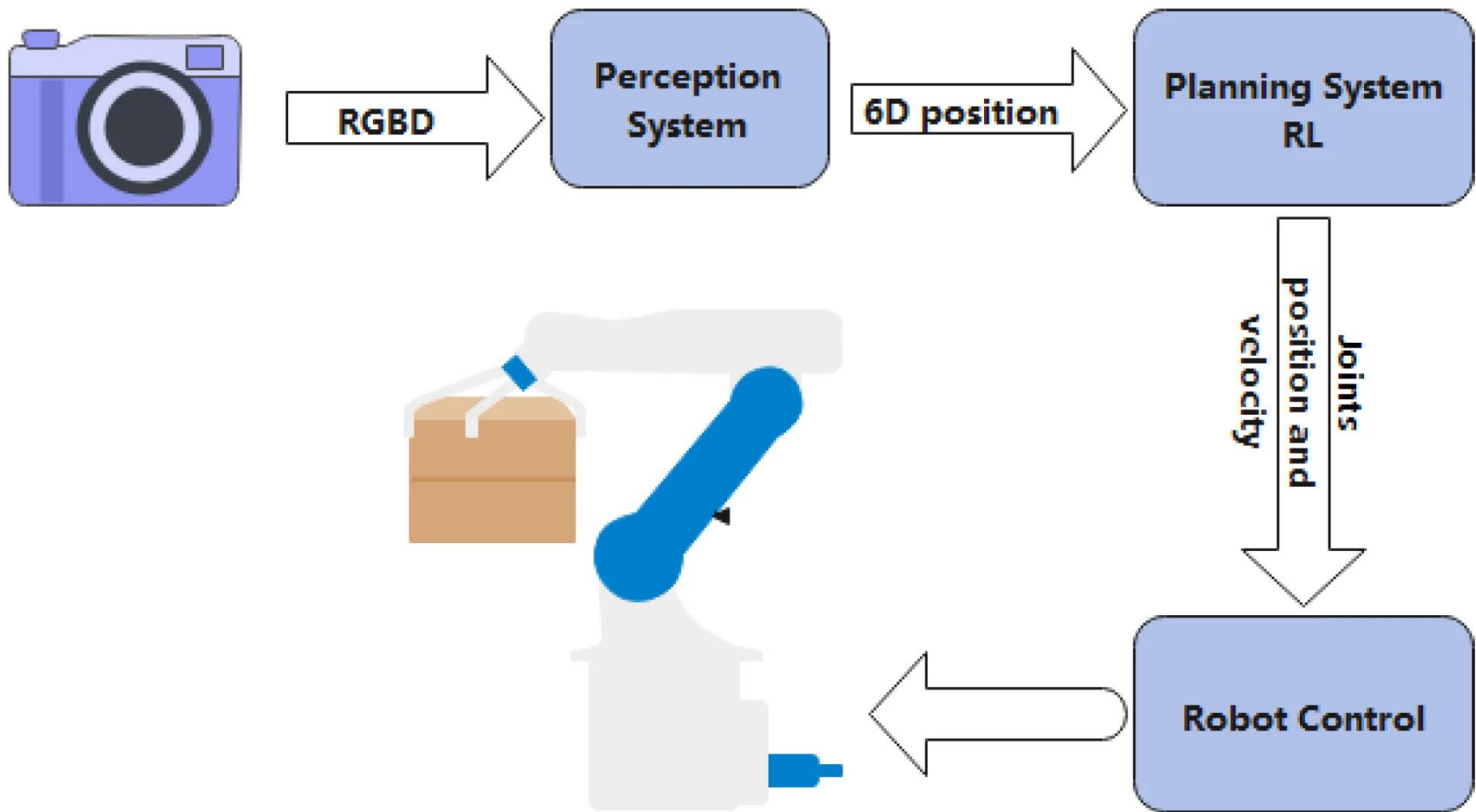
Common Reinforcement Learning Algorithms

- Q-Learning
- SARSA
- Deep Q Networks (DQN)
- Policy Gradient Methods

Applications of Reinforcement Learning

- Game playing (Chess, Video games)
- Robotics
- Self-driving cars
- Recommendation systems
- Resource management
- <https://www.youtube.com/watch?v=jwSbzNHGfIM>

Applications of Reinforcement Learning



Applications of Reinforcement Learning



Advantages of Reinforcement Learning

- Learns from experience
- No labeled data required
- Handles complex environments
- Improves over time

Limitations of Reinforcement Learning

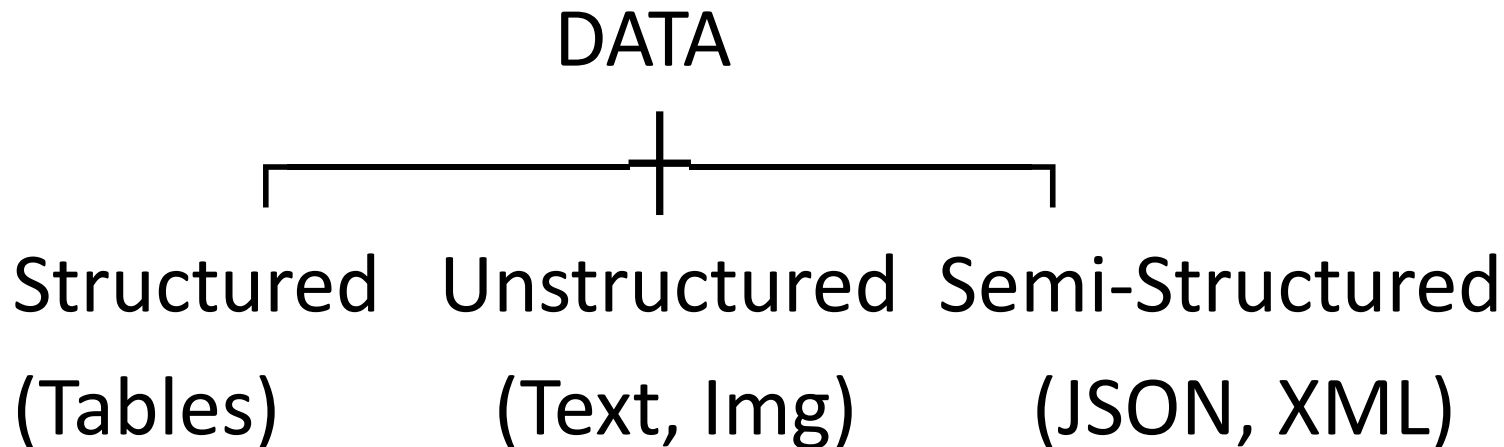
- Requires many interactions
- Training can be slow
- Designing reward function is difficult
- Computationally expensive

Types of Data in Machine Learning

- Structured Data – Tables, databases (rows & columns)
- Unstructured Data – Text, images, audio, video
- Semi-Structured Data – JSON, XML, logs
- Numerical vs Categorical Data
- Time-Series & Spatial Data

Types of Data

- A clean data-type tree:



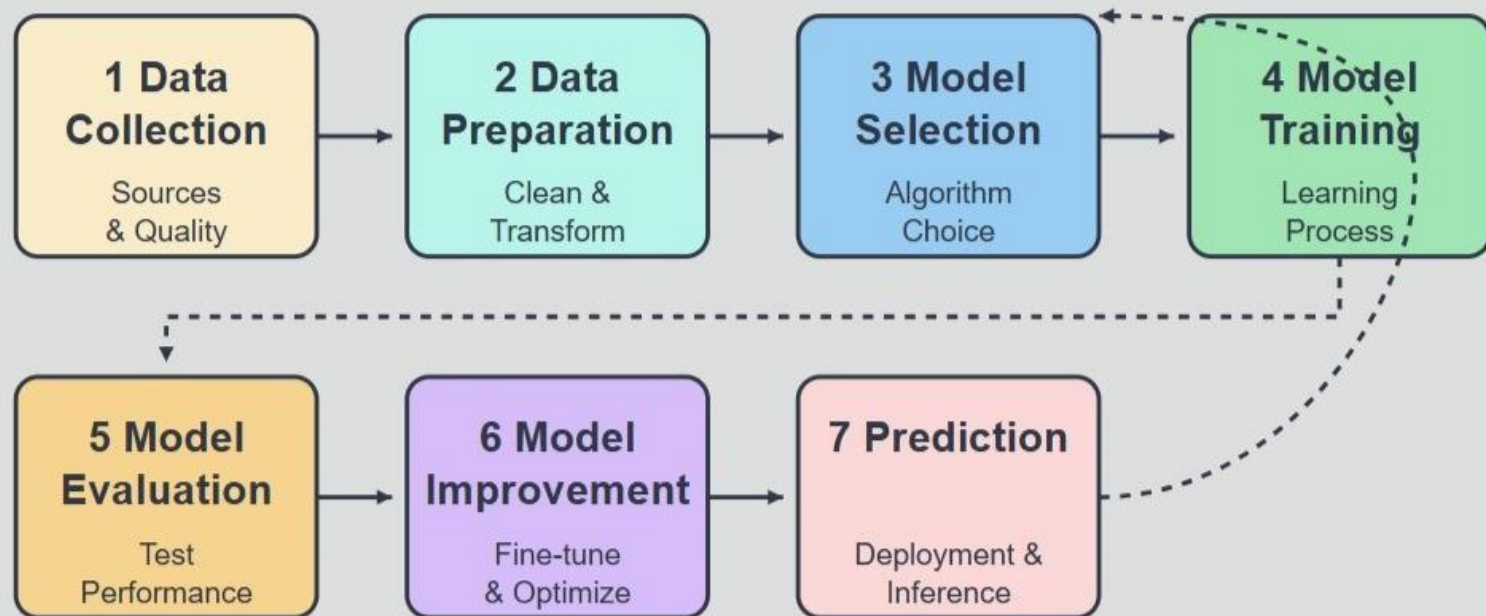
Matching in Machine Learning

- Matching means finding similarity or correspondence between data points
- Exact Matching – String or value equality
- Approximate Matching – Similarity-based (distance measures)
- Feature Matching – Used in images & pattern recognition
- Applications: Recommendation systems, NLP, Image recognition

Input Data → Feature Extraction → Similarity Measure → Best Match

Stages in Machine Learning

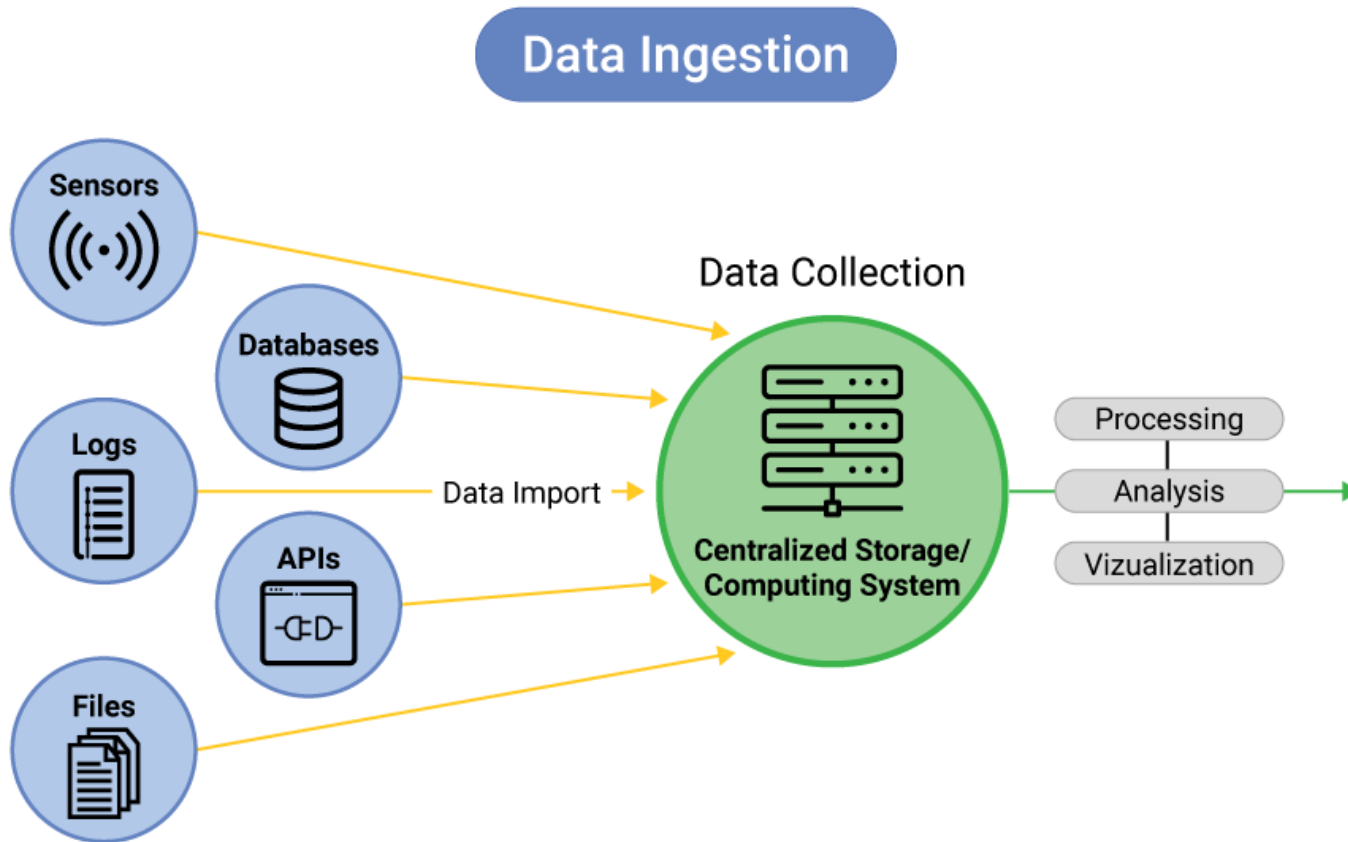
THE 7 STAGES OF MACHINE LEARNING A SIMPLE WORKFLOW



Data Acquisition

- Data Acquisition is the process of **collecting raw data** required for training machine learning models.
- The quality and quantity of data **directly affect model accuracy**.
- Data can be collected from multiple heterogeneous sources.

Data Acquisition



Data Acquisition

- Structured sources: Databases, CSV files
- Unstructured sources: Text, images, videos
- Real-time sources: Sensors, IoT devices

Feature Engineering

- Feature Engineering is the process of **converting raw data into meaningful features.**
- Well-engineered features **improve model performance.**
- It requires both **domain knowledge and analytical skills.**

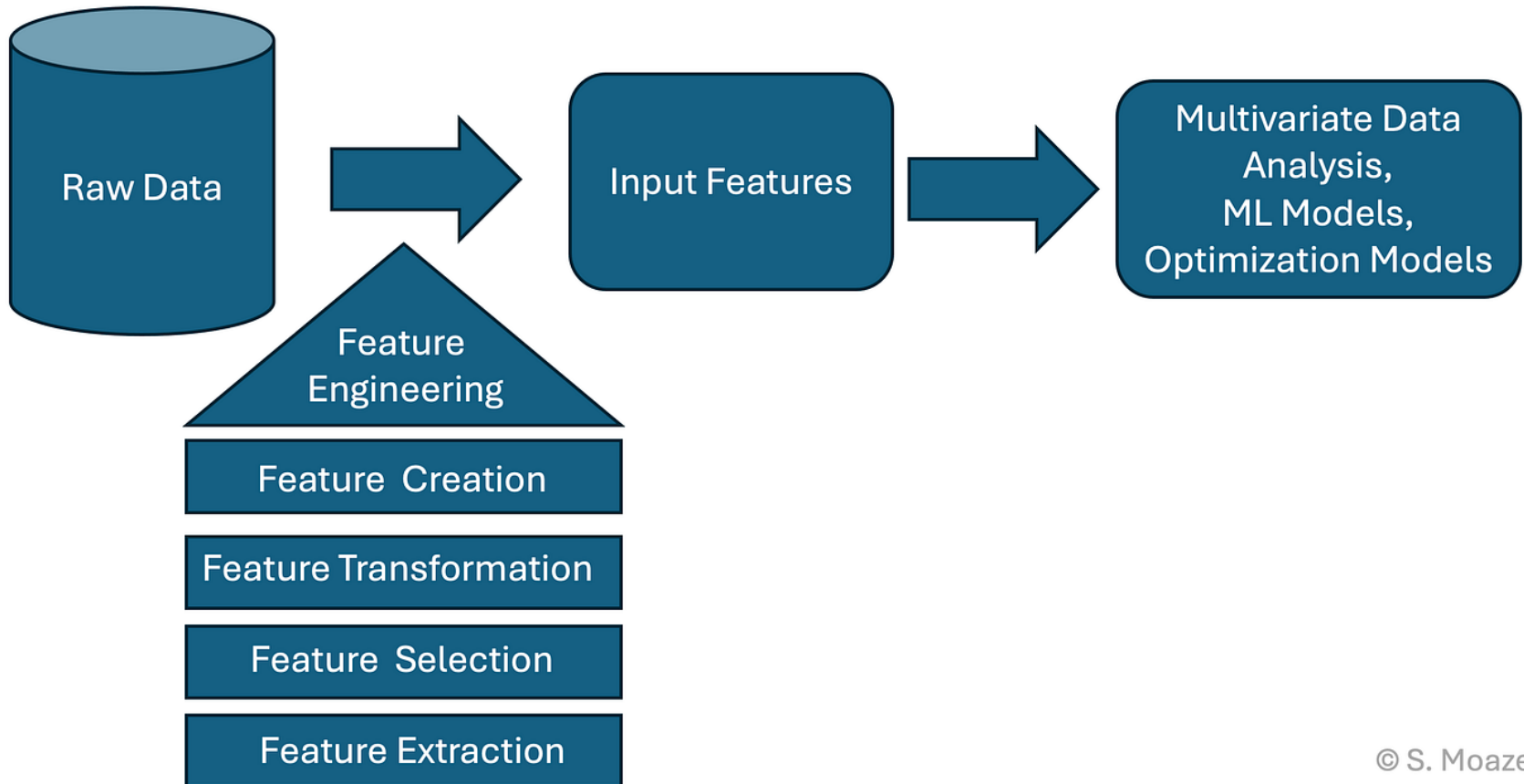
Feature Engineering



Important Steps in Feature Engineering



Feature Engineering



Feature Engineering

- **Examples**
- Feature Extraction: Text → TF-IDF, Images → Edge detection
- Feature Selection: Removing irrelevant attributes
- Feature Transformation: Normalization, scaling, encoding

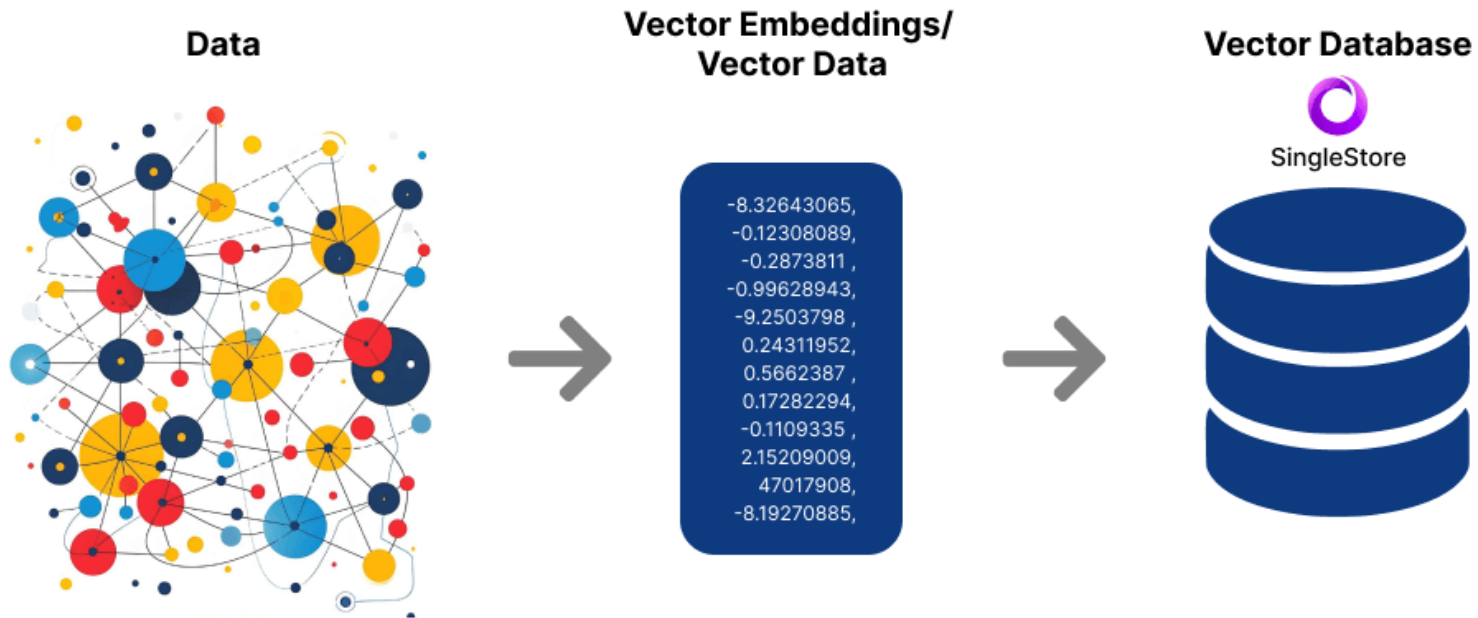
Data Representation

- Data Representation defines **how data is encoded** so ML algorithms can understand it.
- Different data types require different representations.
- Proper representation helps algorithms **learn patterns efficiently**.

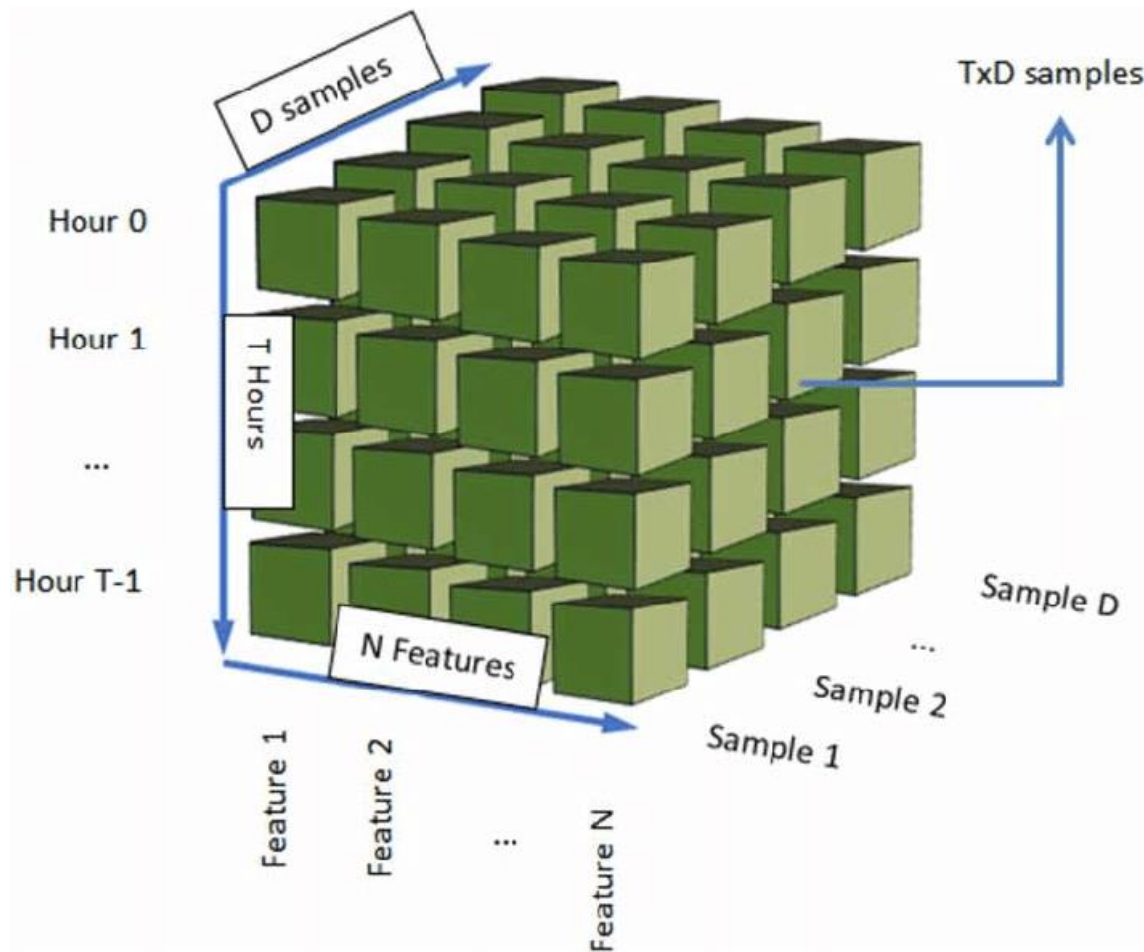
Data Representation

- Tabular Data → Rows & Columns
- Vector Data → Numerical Feature Vectors
- Matrix Data → Image Pixels
- Embeddings → Dense Vector Representations

Data Representation



Data Representation



Data Representation

- **Examples**
- CSV files → Tabular format
- Images → Pixel matrices
- Text → Word embeddings (Word2Vec, BERT)

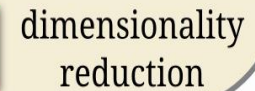
Model Selection

- Definition: Process of choosing the best ML algorithm/model for a given problem.
- Factors: Data type, task (classification/regression), complexity, interpretability.
- Goal: Balance bias-variance tradeoff.

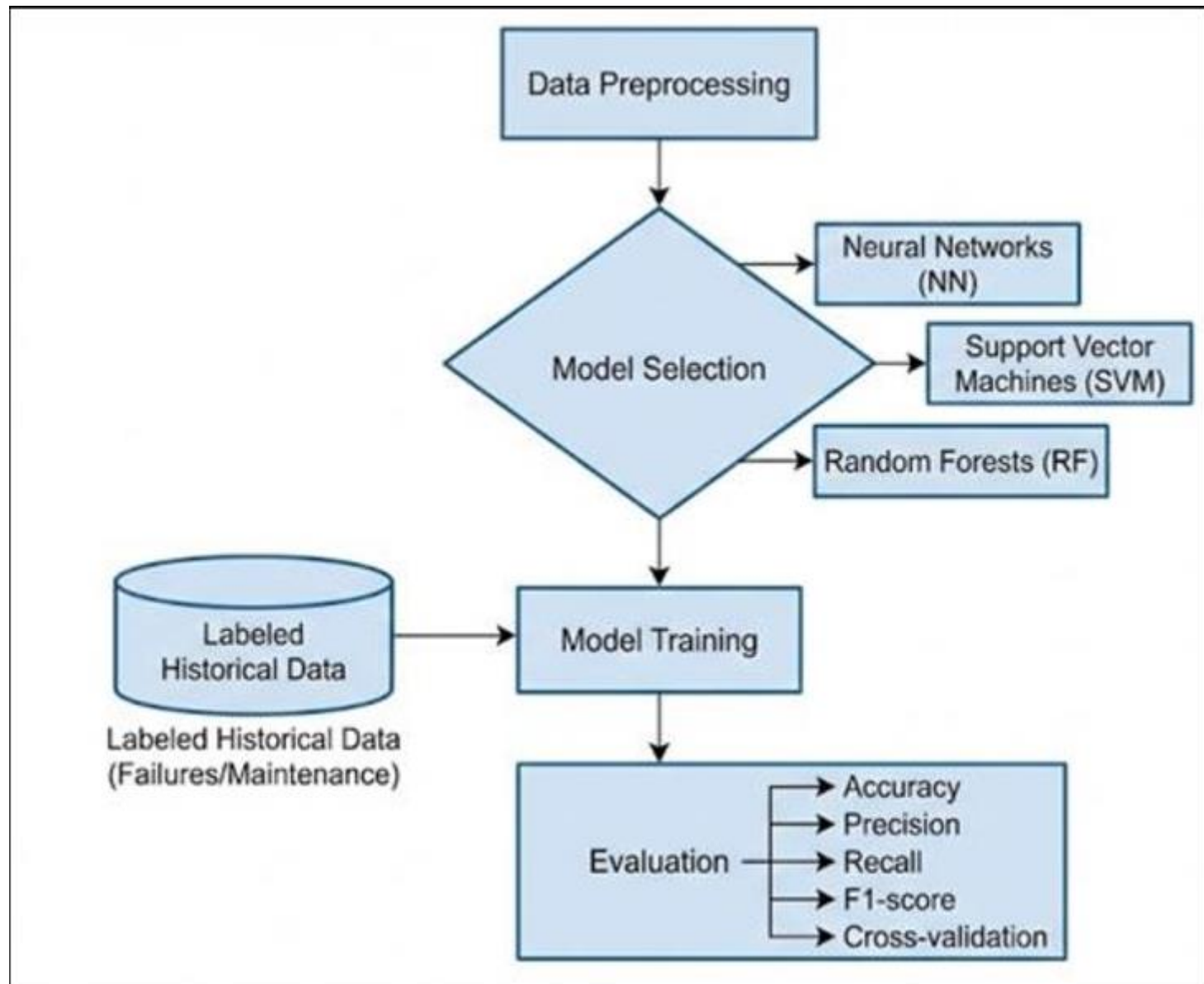
Common Approaches

- Rule-based (e.g., Scikit-learn cheat sheet).
- Try multiple models and compare.
- Considerations: Linear vs. non-linear, ensemble methods.

classification



General Model Selection Flow



Best Practices

- Use cross-validation.
- Avoid overfitting.
- Tools: Scikit-learn, AutoML.

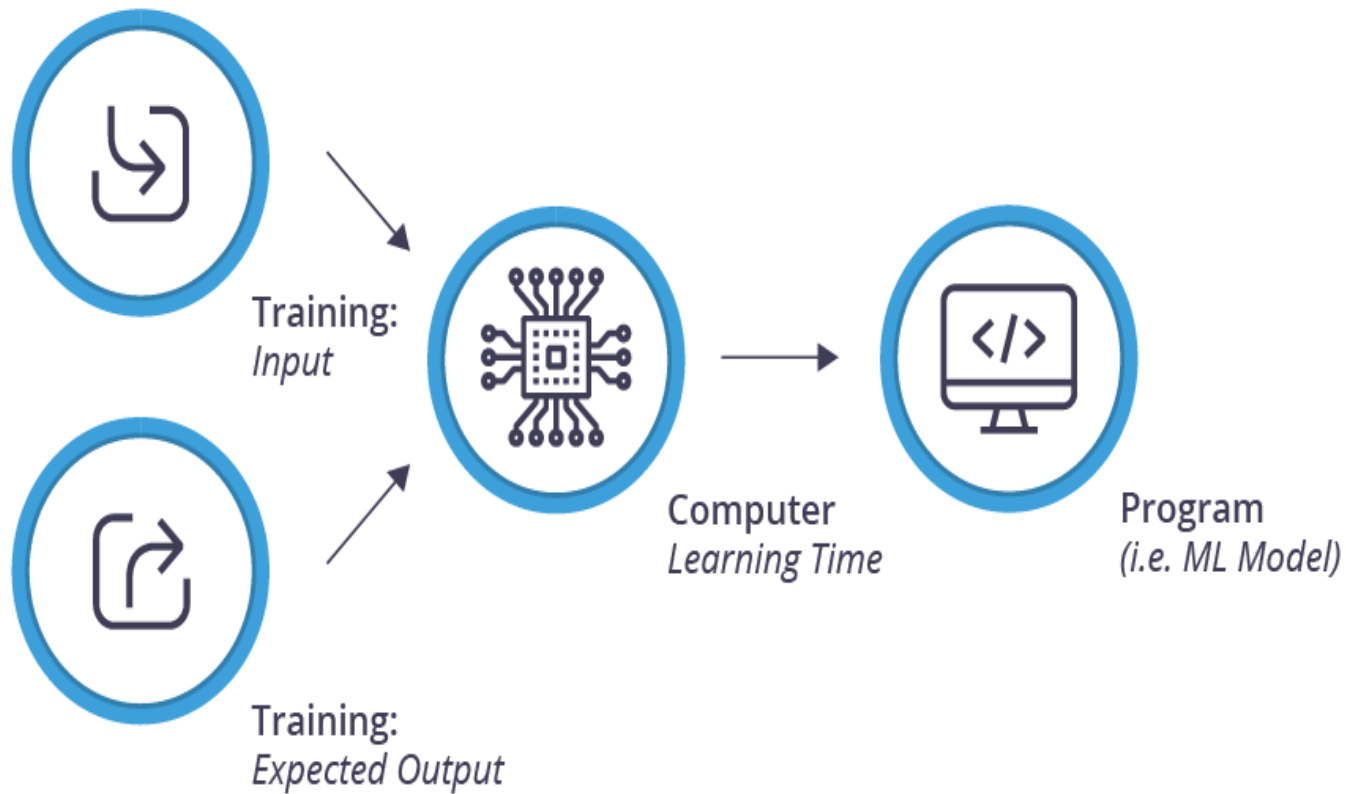
Model Learning

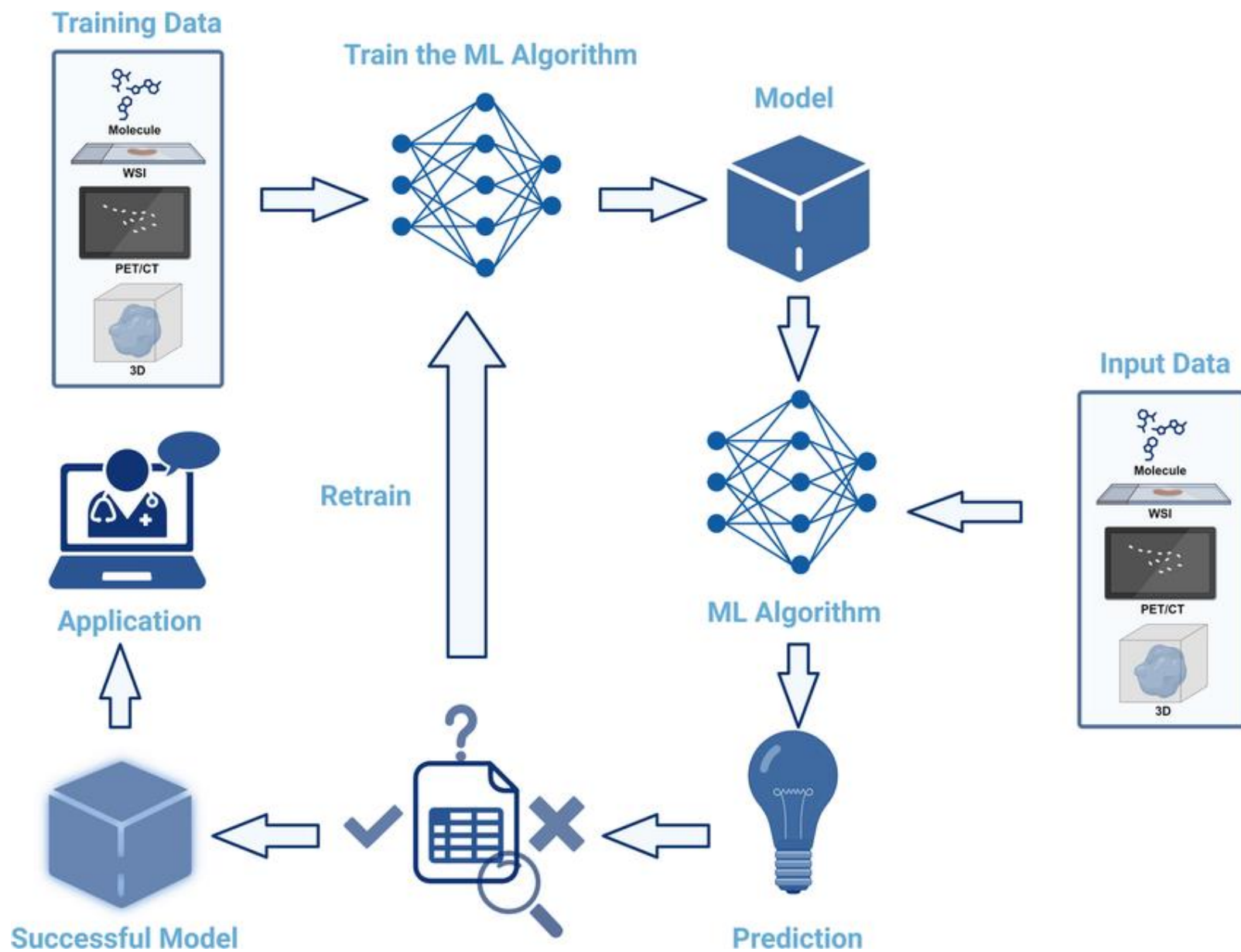
- Also called "training".
- Model adjusts parameters to minimize loss on training data.
- Key steps: Forward pass, loss calculation, backpropagation (for neural nets).

Training Process Steps

- Data preparation → Model initialization → Iteration (epochs) → Optimization (e.g., Gradient Descent).

The Machine Learning **Training** Process





Challenges

- Overfitting/underfitting.
- Use regularization, early stopping.

Overfitting

- **Overfitting** occurs when a machine learning model learns the training data **too well**, including its noise and outliers, and as a result **performs poorly on new, unseen data**.

Why Overfitting Happens

- **Too complex model** (many parameters, deep networks)
- **Small training dataset**
- **Too many features**
- **Training for too many epochs**
- **Noise in data**

Underfitting

- **Underfitting** occurs when a machine learning model is **too simple** to capture the underlying patterns in the data, resulting in **poor performance on both training and testing data**.

Why Underfitting Happens

- **Model is overly simple** (e.g., linear model for non-linear data)
- **Insufficient features**
- **Excessive regularization**
- **Insufficient training time**
- **Poor feature engineering**

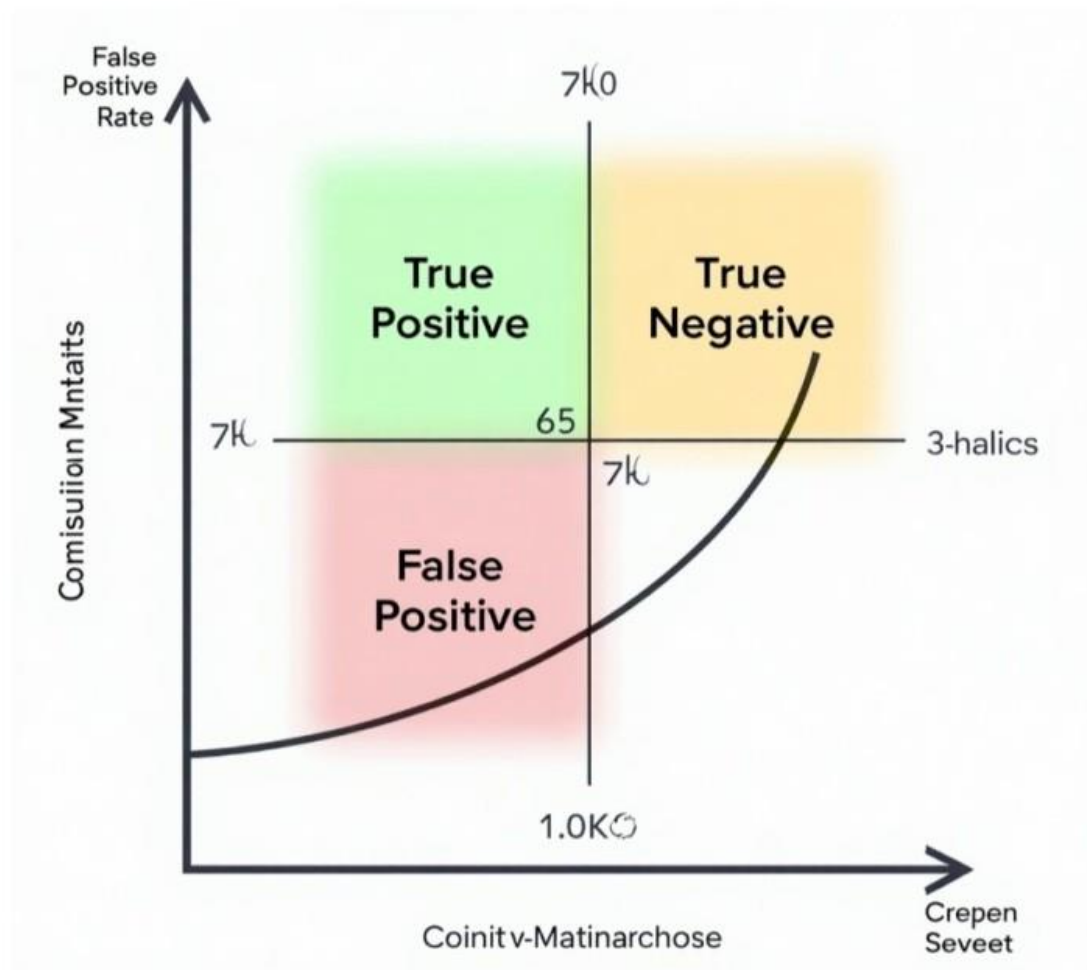
Model Evaluation

- **Why Evaluate Models?**
- Assess generalization.
- Compare models.
- Metrics depend on task (classification, regression).

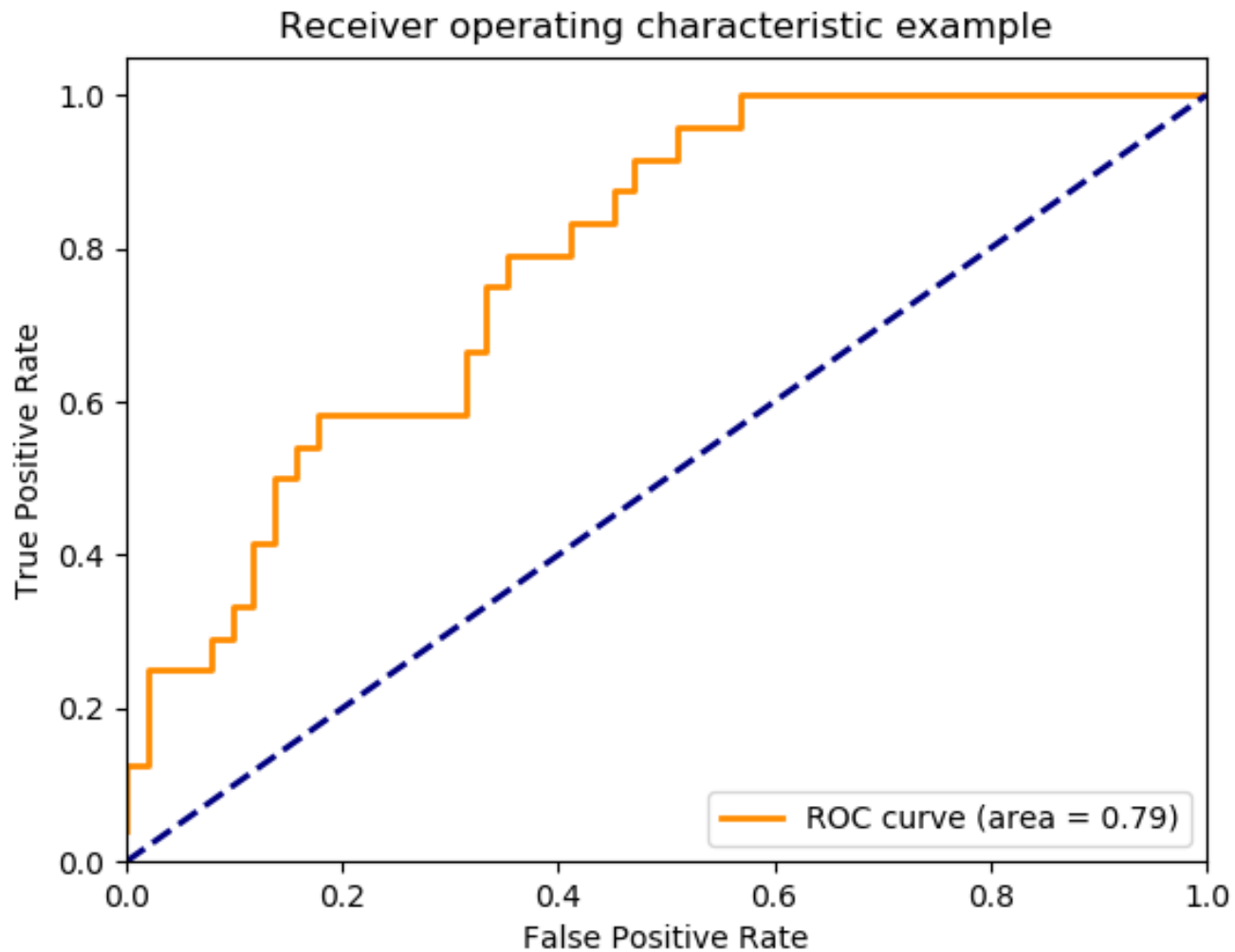
- **Key Metrics**

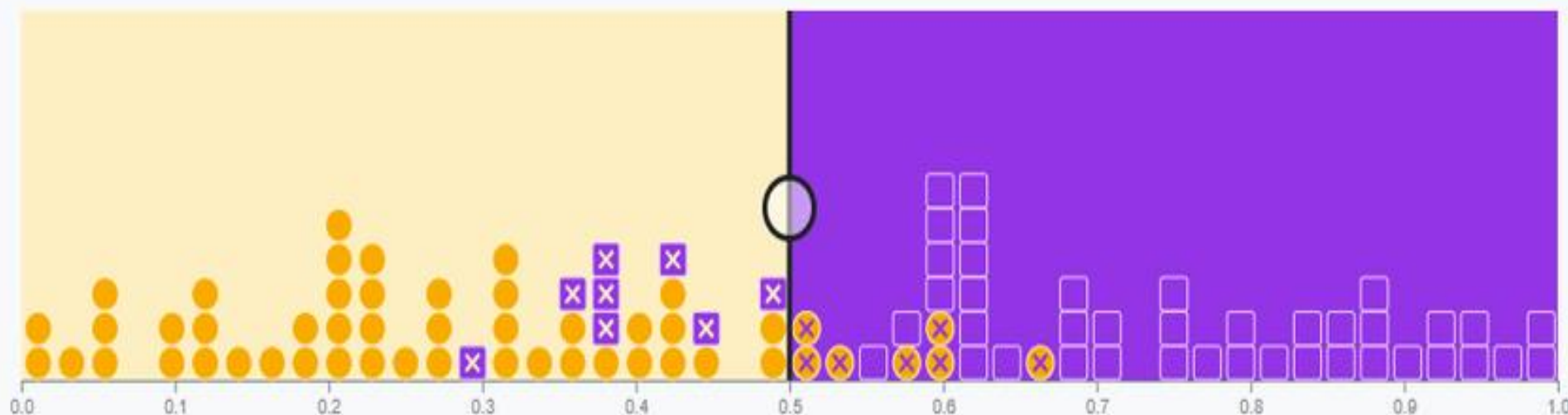
- Classification: Accuracy, Precision, Recall, F1, ROC-AUC.
- Regression: MSE, RMSE, R^2 .

Confusion Matrix & Metrics



ROC Curve and AUC





Threshold: 0.50

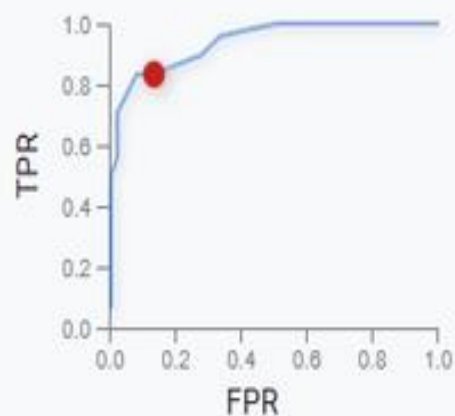
Confusion matrix

	Actually positive	Actually negative
Predicted positive	□ TP=40	⊗ FP=7
Predicted negative	× FN=8	● TN=44

Metrics

Accuracy 0.85
Precision 0.85
Recall 0.83

ROC curve



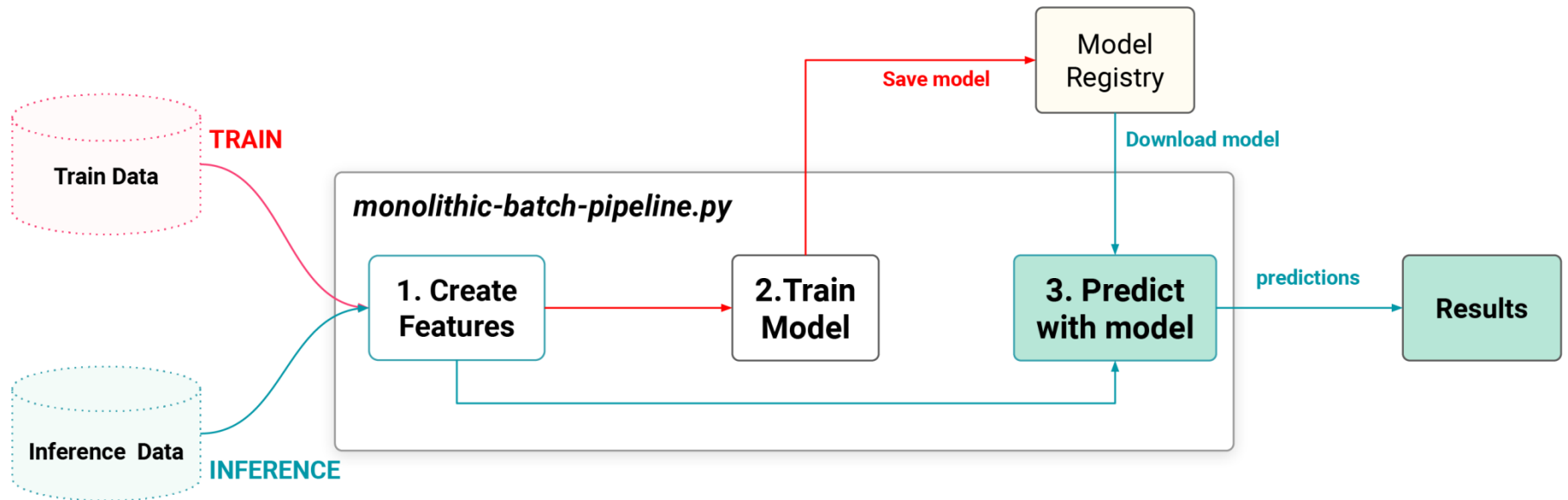
- **Advanced Techniques**
- Cross-validation, hold-out testing.

Model Prediction

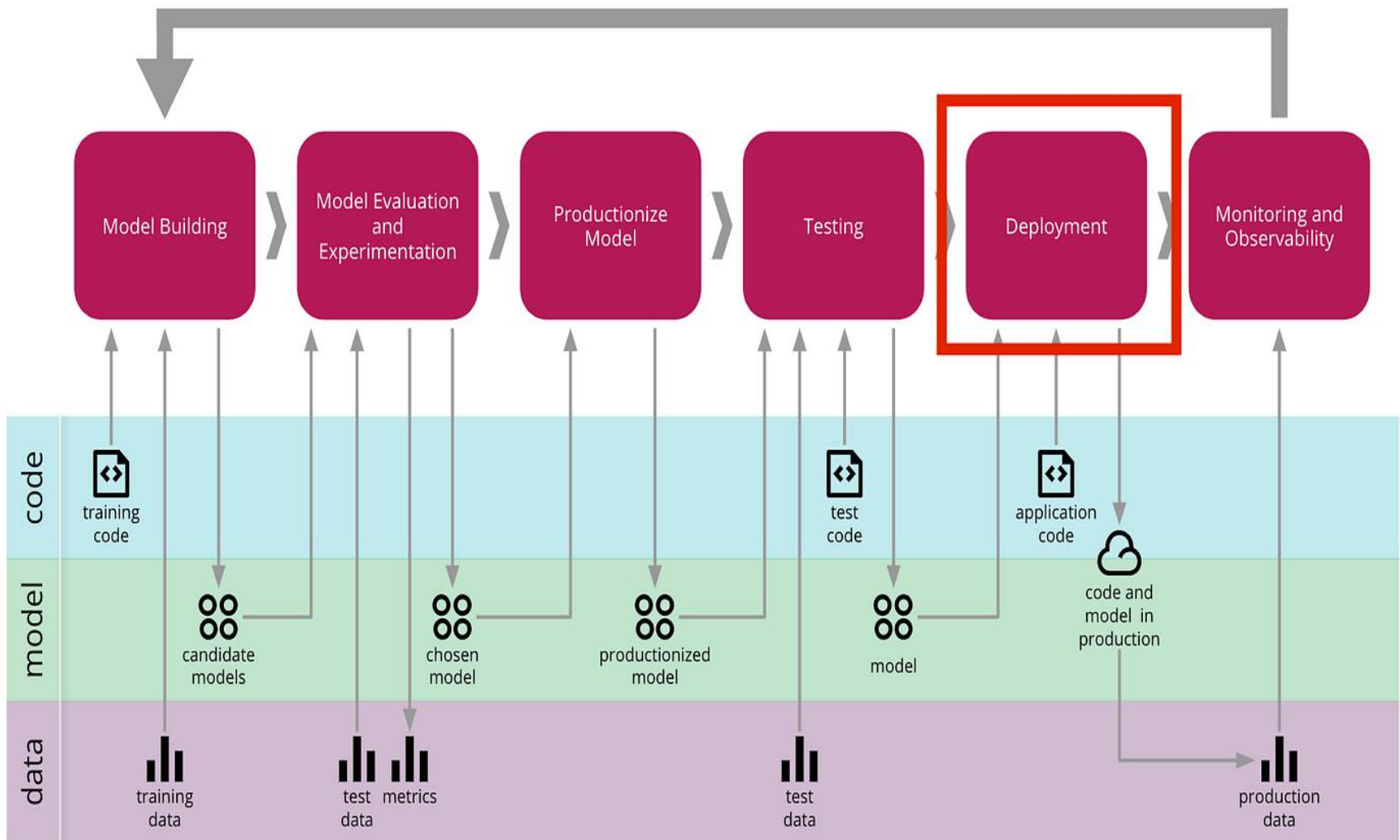
- **Prediction vs. Training**
- Training: Learn from data.
- Prediction (Inference): Apply trained model to new data.

- **Inference Pipeline**
- Preprocess input → Load model → Generate output → Post-process.
- Deployment: Batch vs. real-time.

Training/Inference Pipelines



Inference Service Architecture



Challenges

- Latency, scalability, drift.
- Tools: TensorFlow Serving, ONNX.
- **Latency** is the **time taken by a system to respond** after receiving an input.
- **In Machine Learning**
- It refers to the **delay between sending data to a model and receiving a prediction.**
- **Example**
- Voice assistant taking time to respond
- Online recommendation system showing suggestions slowly

- **Scalability** is the ability of a system to **handle increased workload** efficiently.
- **In Machine Learning**
- It means the system can manage:
- More users
- More data
- More prediction requests
- without performance degradation.
- **Example**
- A model serving 1,000 users today and 1 million users tomorrow
- Training on GBs → TBs of data

- **Drift** occurs when the **data or relationships change over time**, causing model performance to degrade.
- **Types of Drift**
- **Data Drift**
Input data distribution changes
 - Example: Customer behavior changes after a festival season
- **Concept Drift**
Relationship between input and output changes
 - Example: Spam patterns evolving over time
- **Why it matters**
- Model accuracy decreases
- Predictions become unreliable
- **Solutions**
- Monitor model performance
- Retrain models periodically
- Use adaptive learning

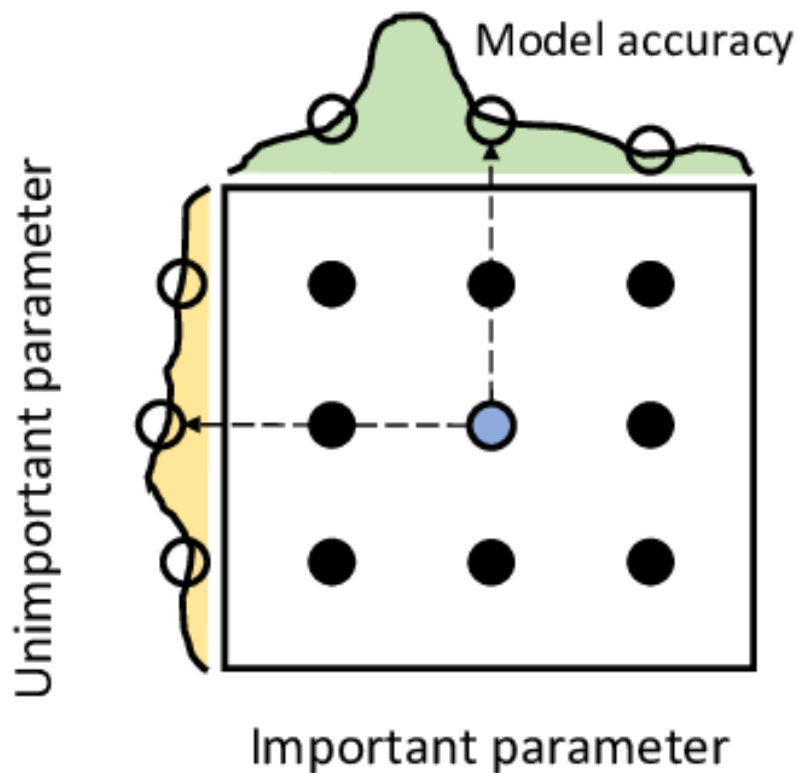
Search and Learning

- **What is Hyperparameter Search?**
- Hyperparameters:
- **Hyperparameters** are **configuration settings** of a machine learning model that are **set before training** and control how the model learns from data (e.g., learning rate, depth).
- **Search:** Find optimal values to improve model performance.

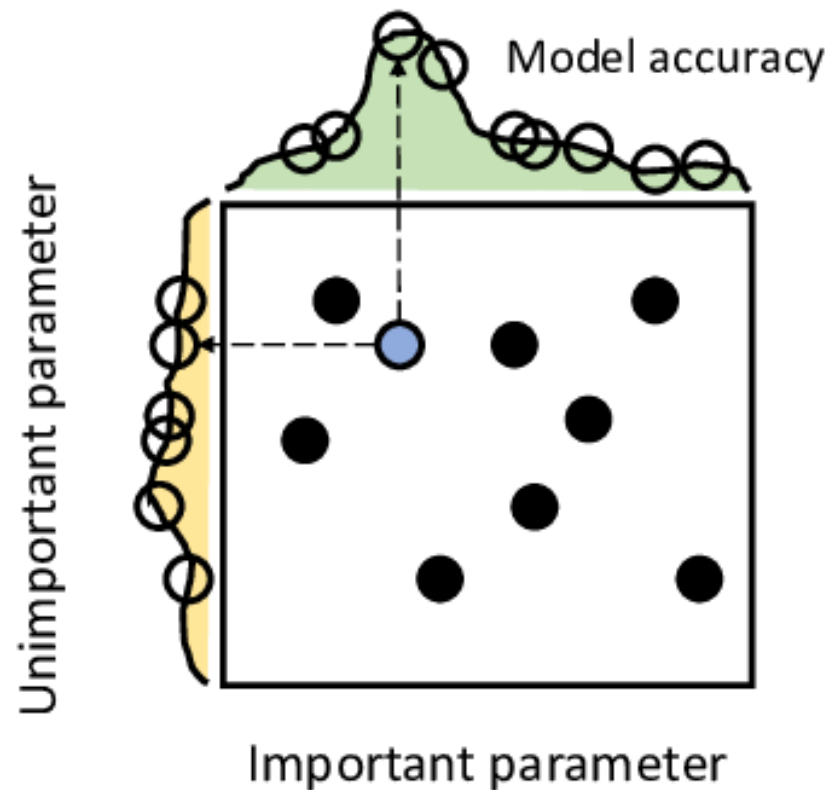
Methods

- Grid Search: Exhaustive.
- Random Search: Efficient.
- Advanced: Bayesian Optimization.

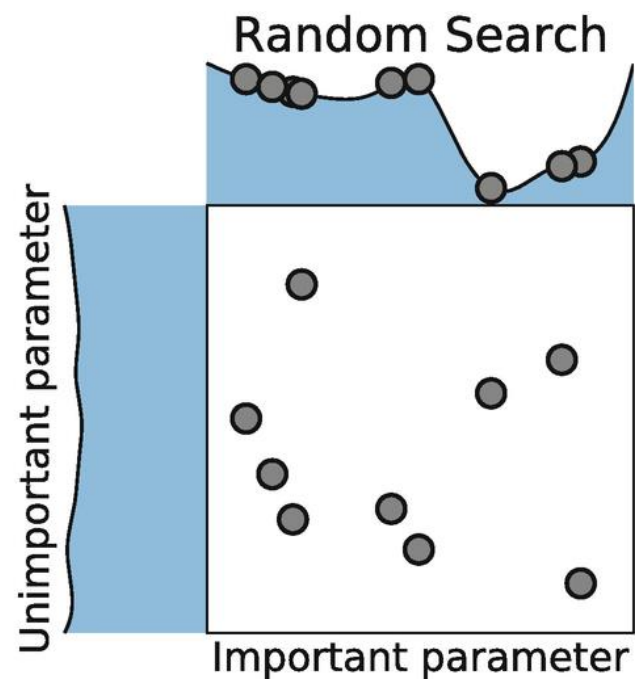
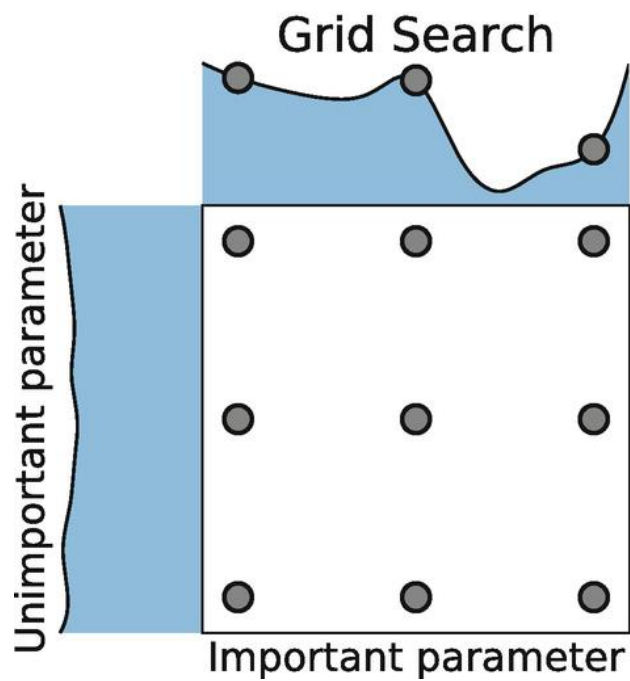
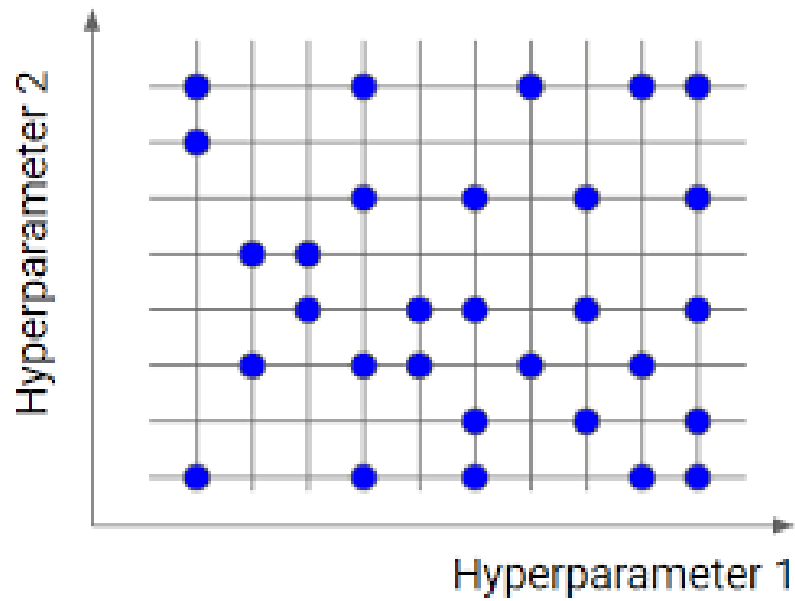
Grid vs. Random Search



(a)



(b)



- **Tools & Best Practices**
- Scikit-learn, Optuna, Ray Tune.

Data Sets

- Importance of Data Splitting
- Prevent overfitting.
- Train: Learn patterns.
- Validation: Tune.
- Test: Final evaluation.

- **Common Splits**
- 70/15/15 or 80/10/10.
- Stratified for imbalanced data.

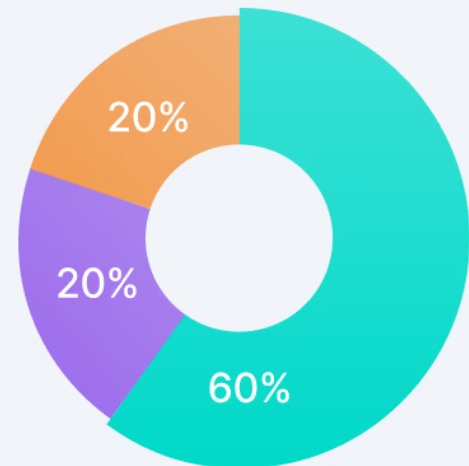
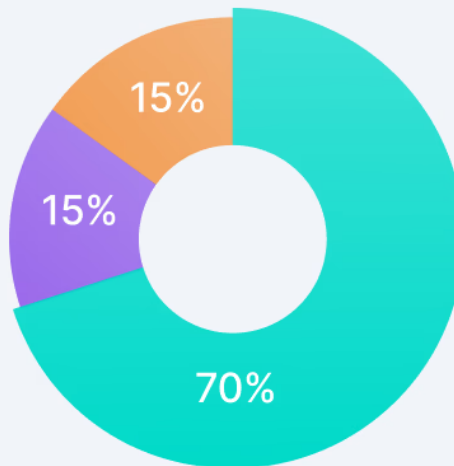
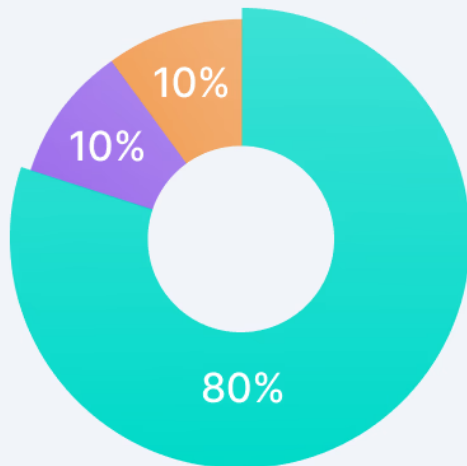
Train/Validation/Test Split

Data Training Needs

Training data

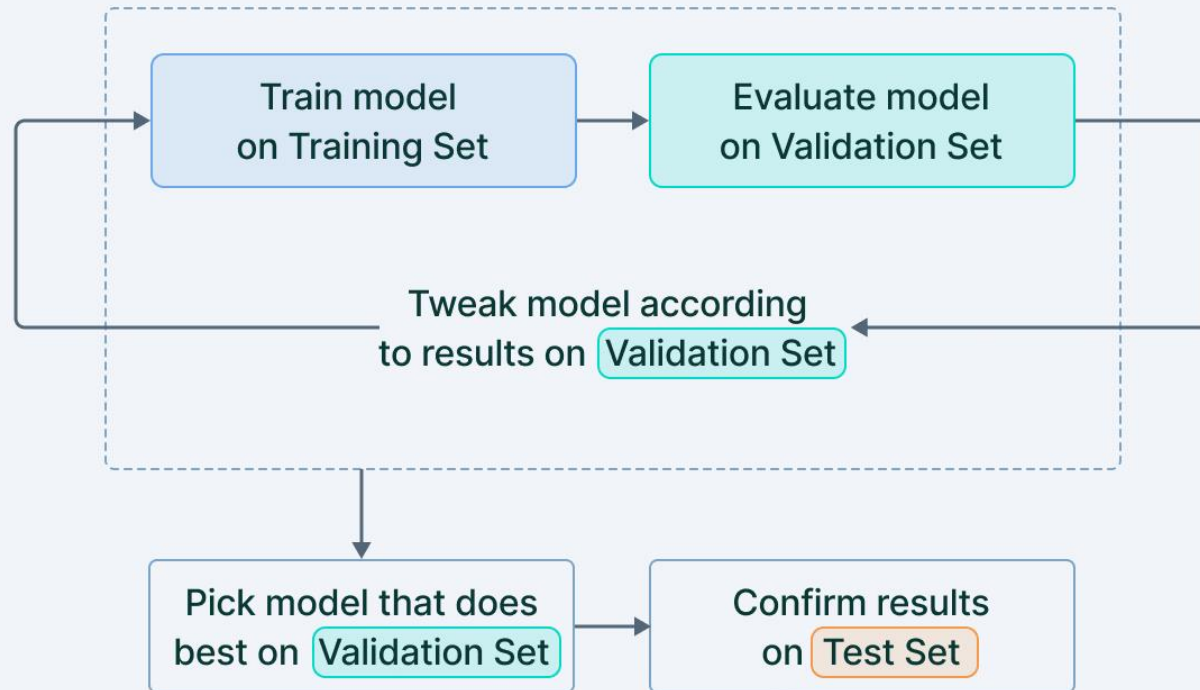
Validation data

Test data



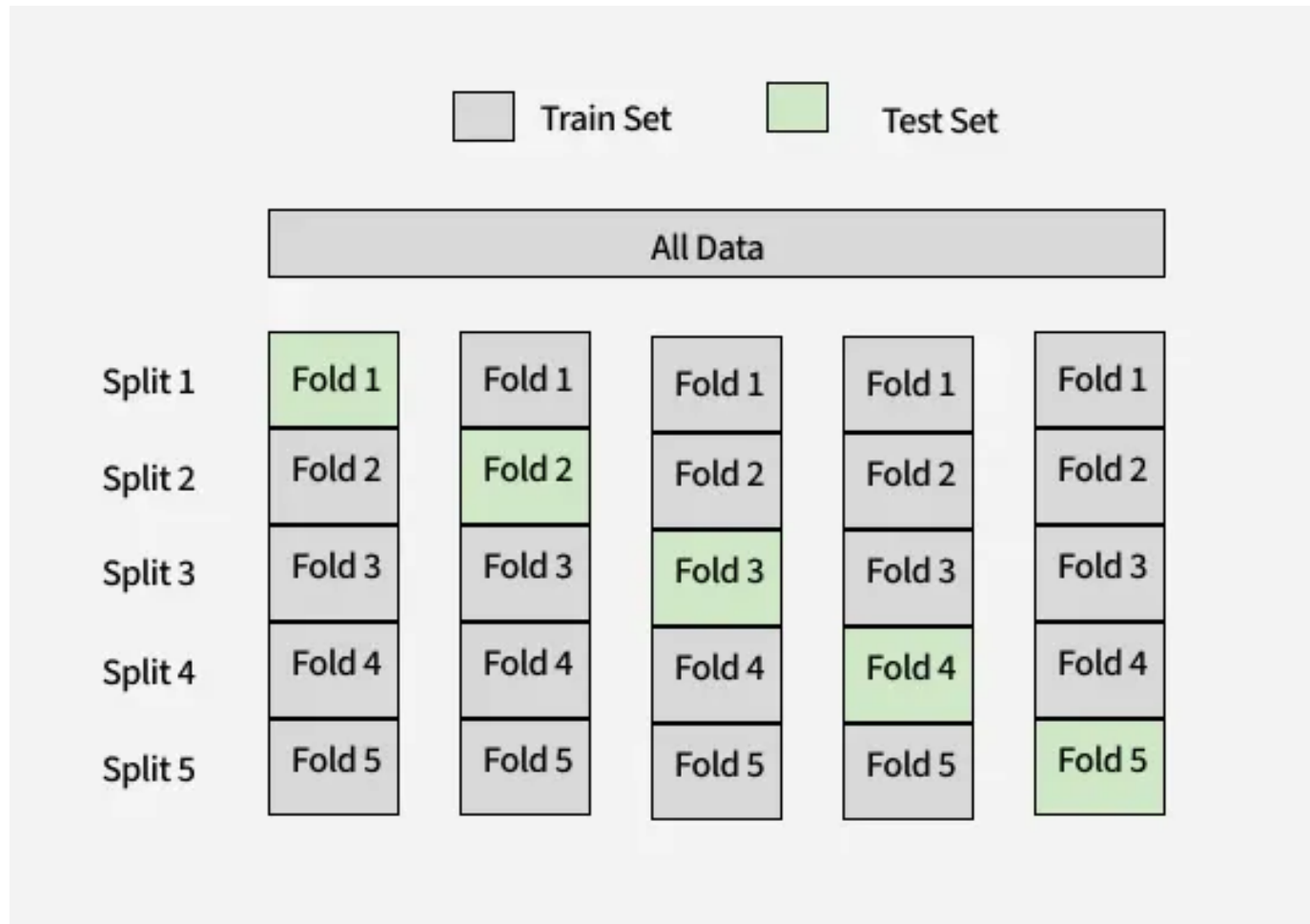
Train/Validation/Test Split

Training data/validation/test

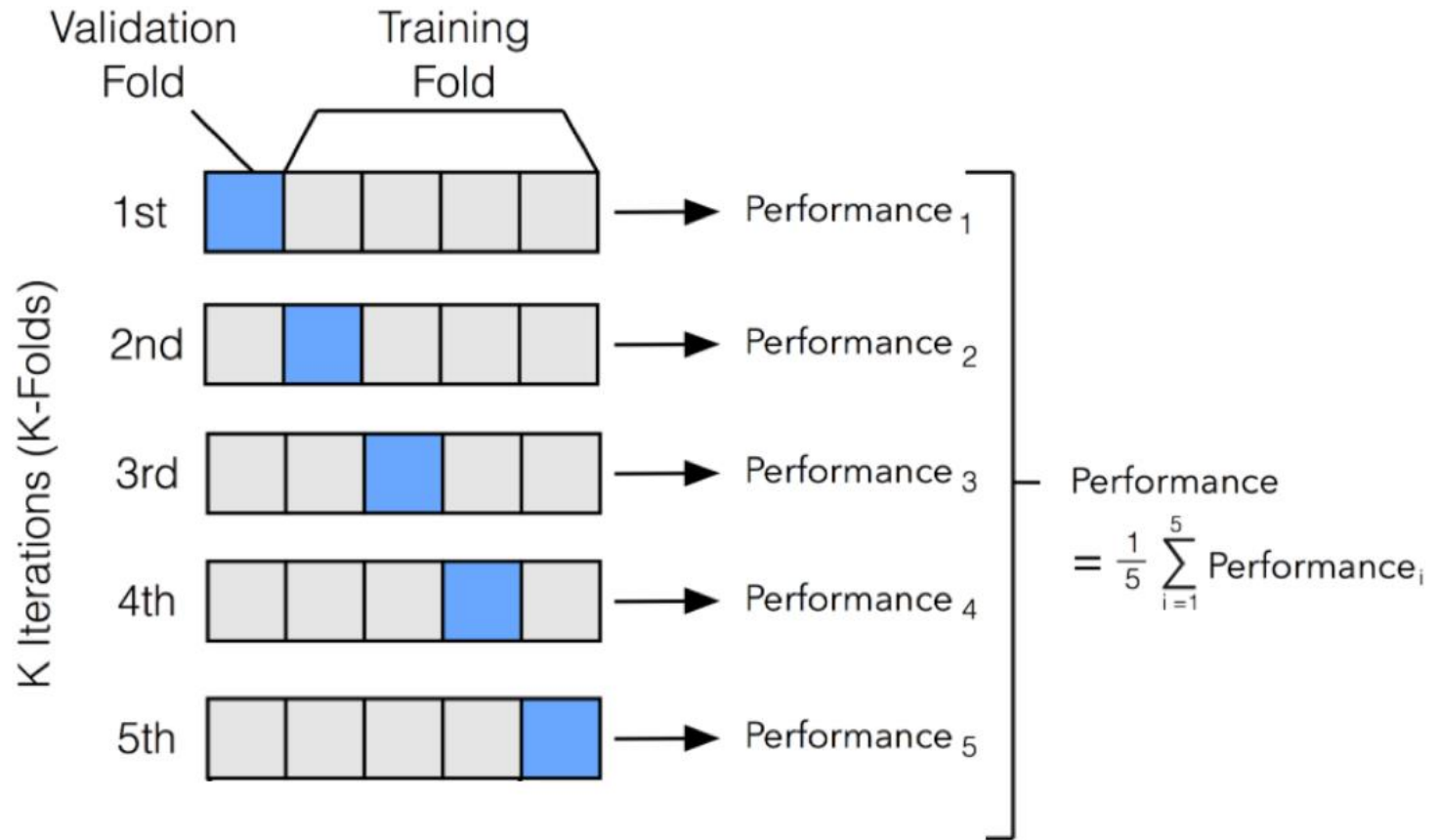


Cross-Validation Diagram

K-Fold for better estimation.



Cross-Validation Diagram



Quaries?